



DIGITAL IMAGE PROCESSING

Lecture 2

Instructor: Sara Tehsin





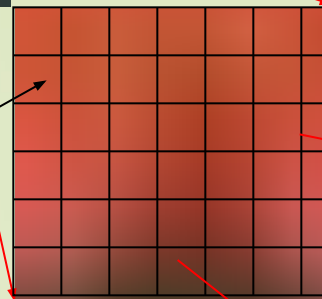
Chapter 2: Digital Image Fundamental



Digital Image



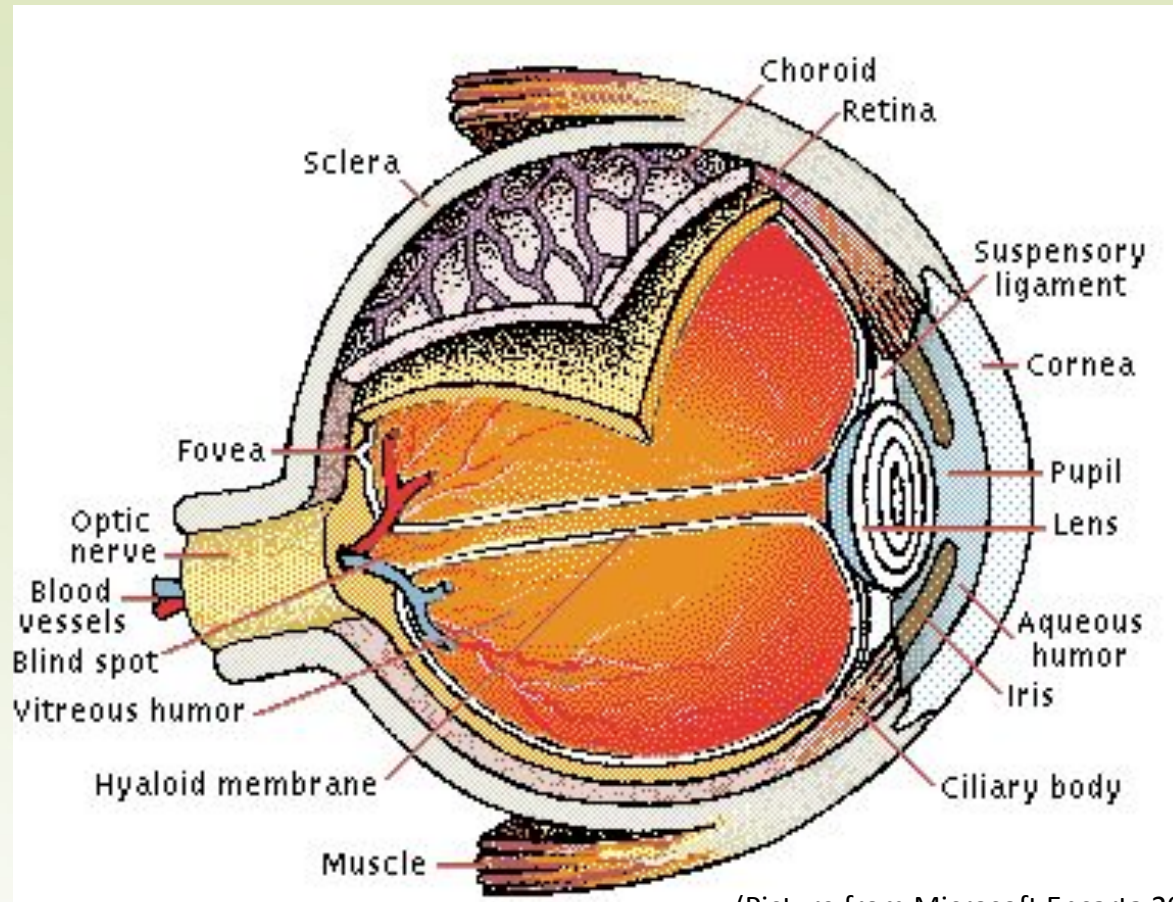
Digital image = a multidimensional array of numbers (such as intensity image) or vectors (such as color image)



Each component in the image called pixel associates with the pixel value (a single number in the case of intensity images or a vector in the case of color images).

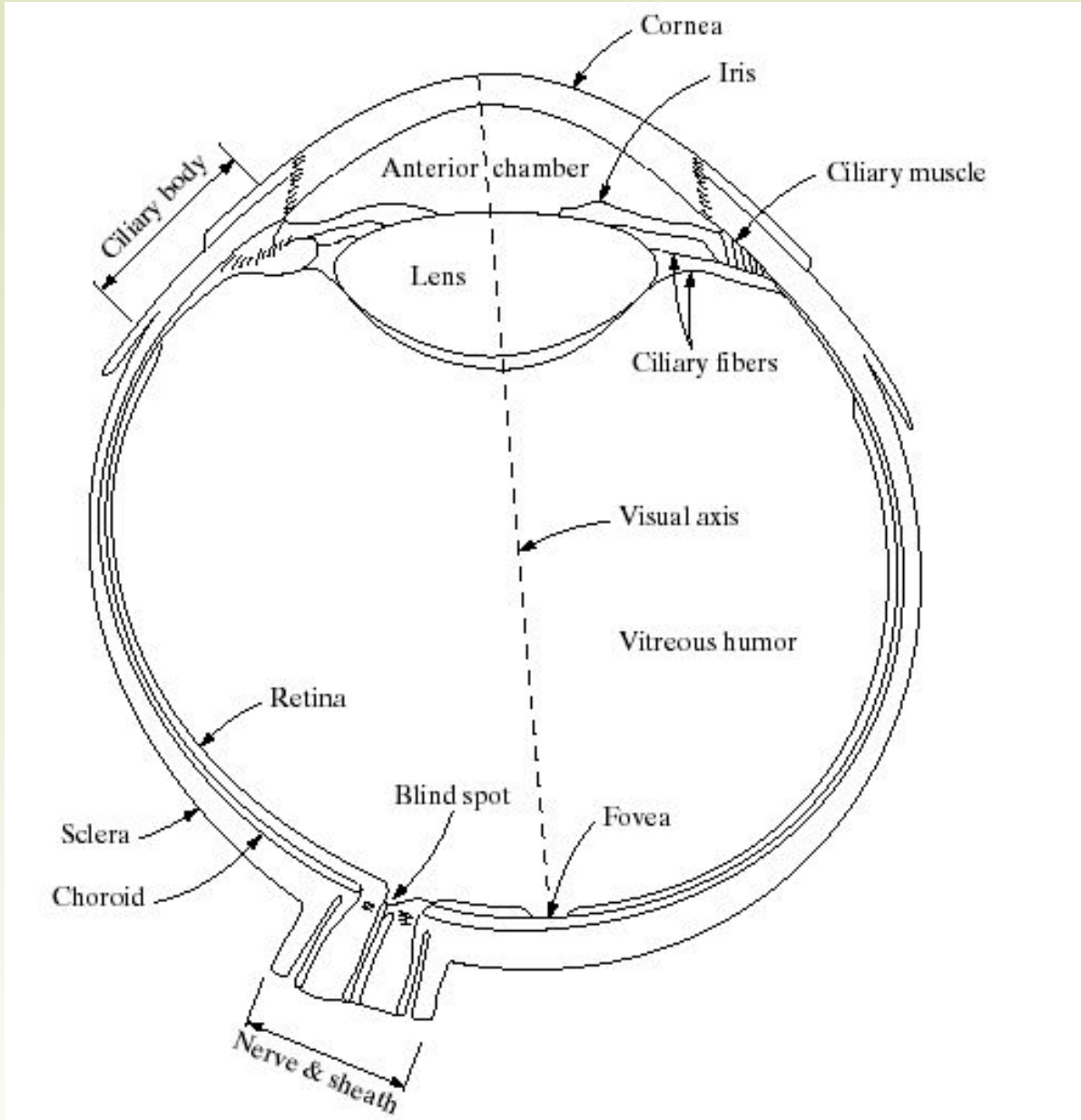
10	10	16	28		
9	65	70	56	43	
15	32	99	70	56	78
32	21	60	90	96	67
	54	85	85	43	92
		32	65	87	99

Visual Perception: Human Eye



(Picture from Microsoft Encarta 2000)

Cross Section of the Human Eye



Visual Perception: Human Eye (cont.)

1. The *lens* contains 60-70% water, 6% of fat.
2. The *iris* diaphragm controls amount of light that enters the eye.
3. *Light receptors* in the *retina*
 - About 6-7 millions *cones* for bright light vision called *photopic*
 - Density of cones is about 150,000 elements/mm².
 - Cones involve in color vision.
 - Cones are concentrated in *fovea* about 1.5x1.5 mm².
 - About 75-150 millions *rods* for dim light vision called *scotopic*
 - Rods are sensitive to low level of light and are not involved color vision.
4. *Blind spot* is the region of emergence of the optic nerve from the eye.

Distribution of Rods and Cones in the Retina

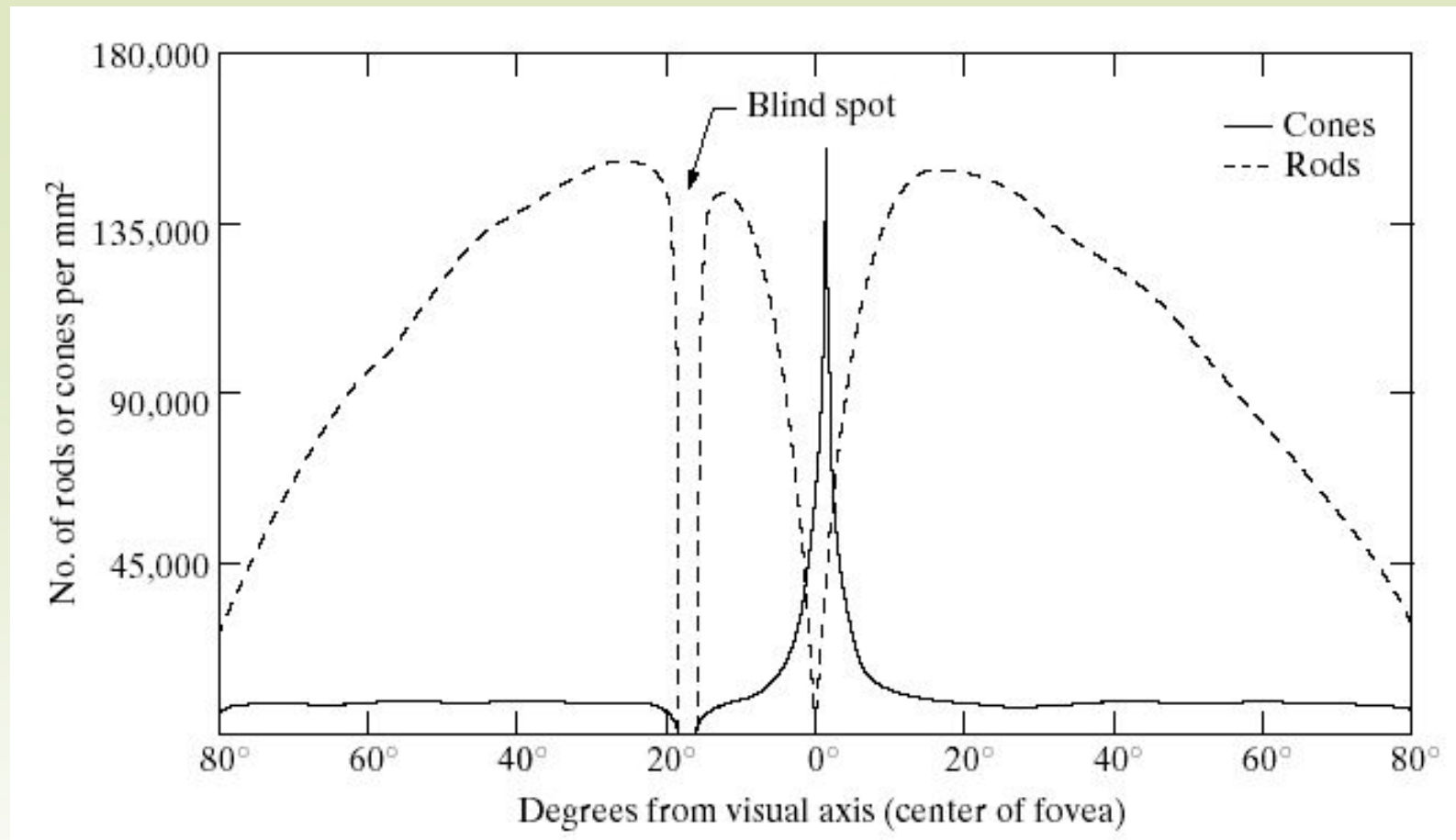
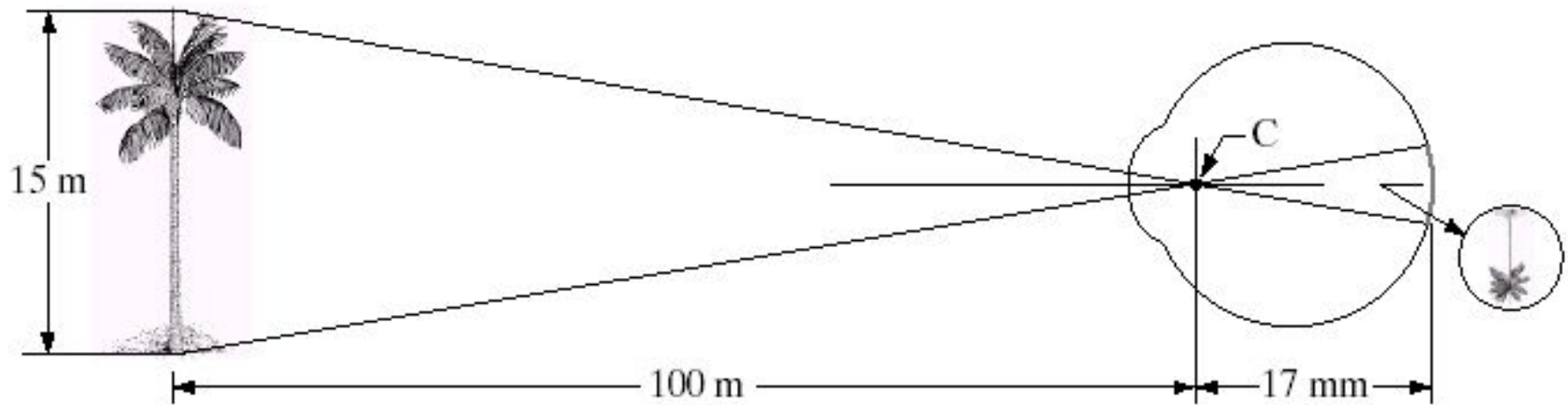
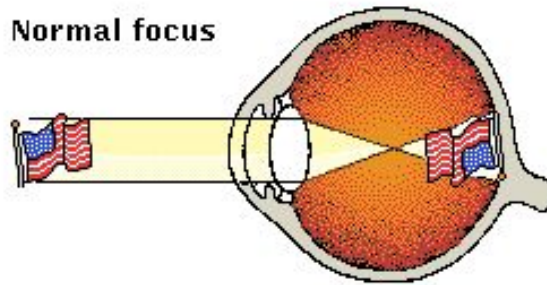


Image Formation in the Human Eye

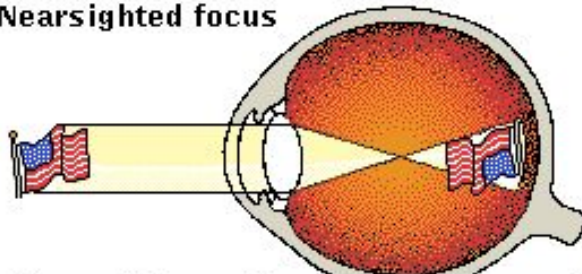


(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Normal focus

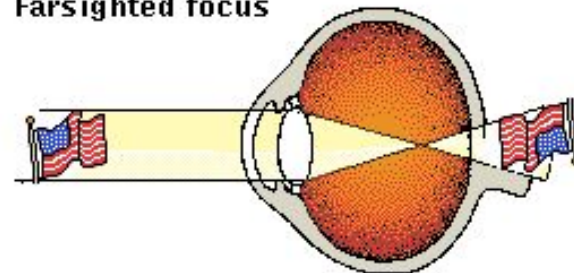


Nearsighted focus



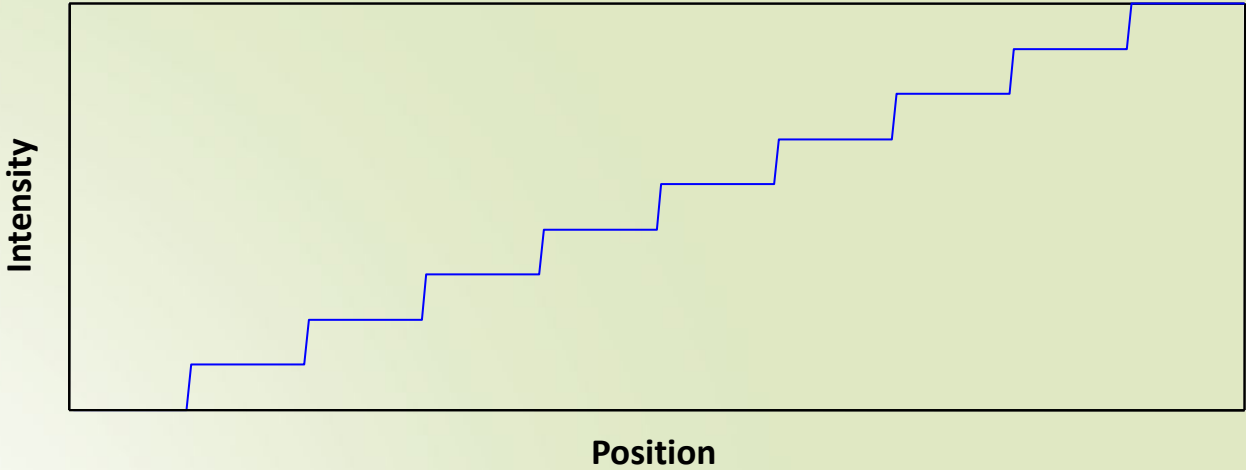
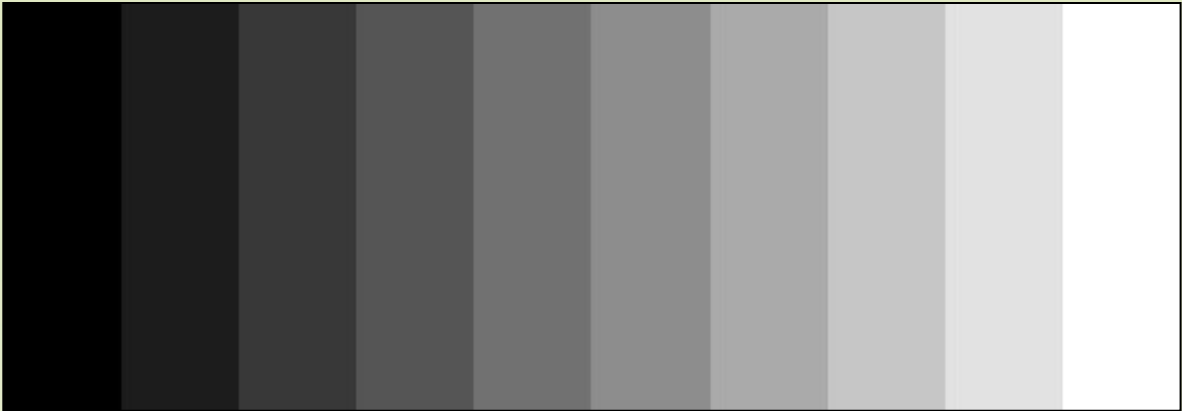
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Farsighted focus

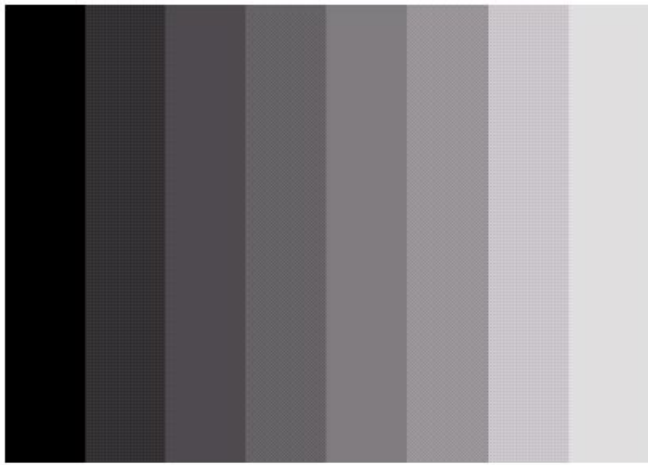


(Picture from Microsoft Encarta 2000)

Brightness Adaptation of Human Eye : Mach Band Effect

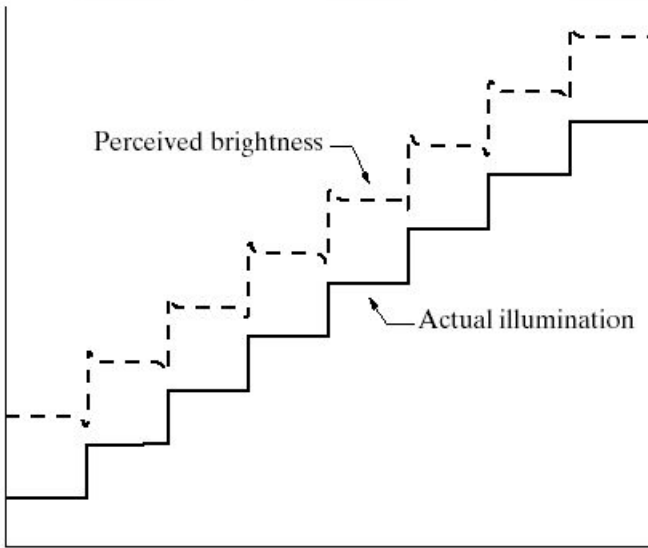


Mach Band Effect

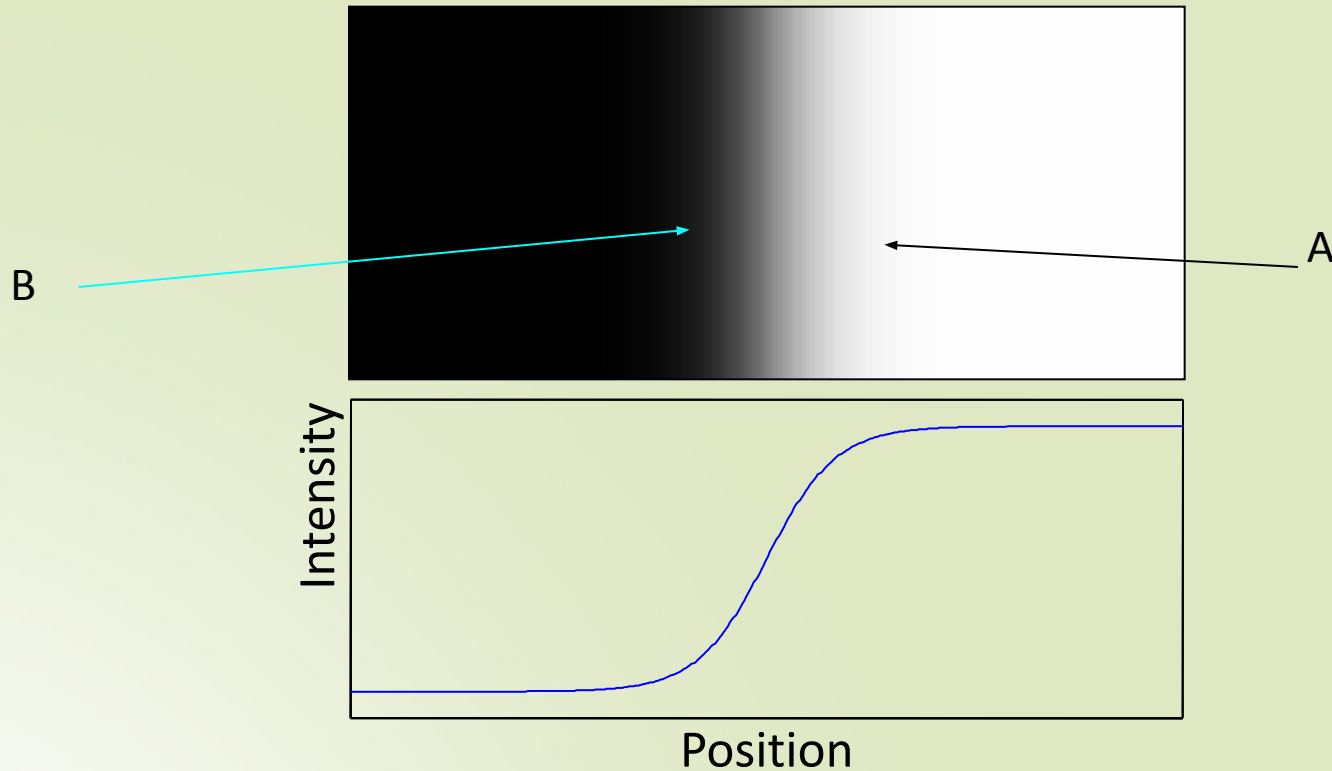


Intensities of surrounding points effect perceived brightness at each point.

In this image, edges between bars appear brighter on the right side and darker on the left side.

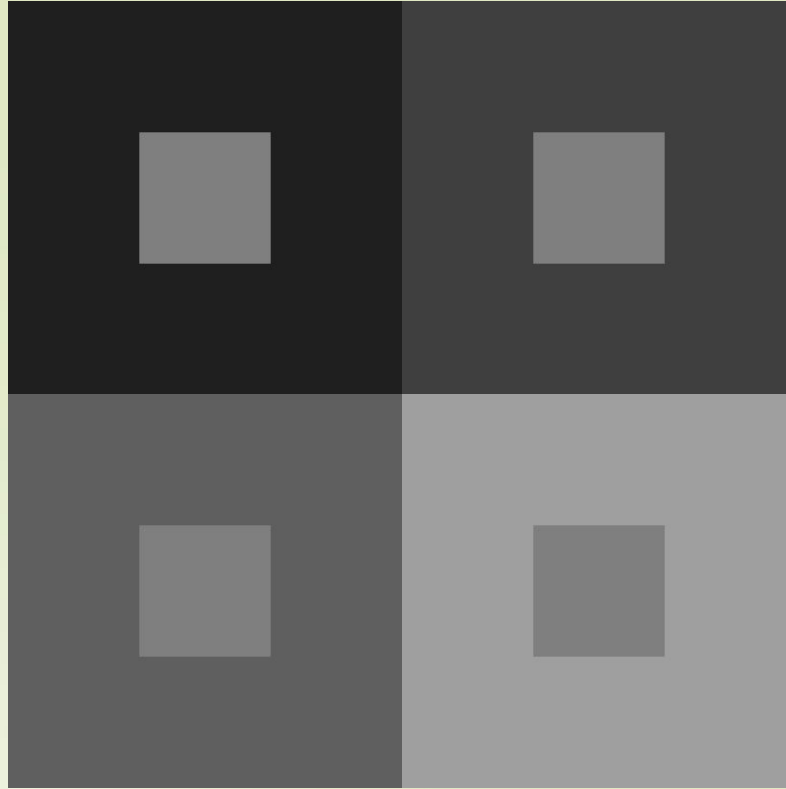


Mach Band Effect (Cont)



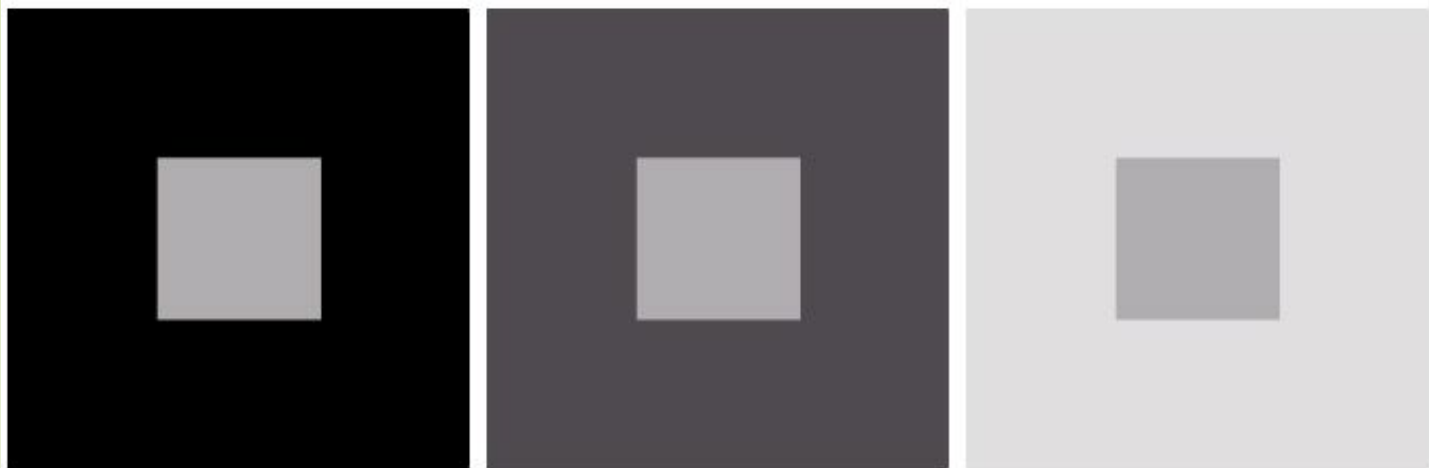
In area A, brightness perceived is darker while in area B is brighter. This phenomenon is called ***Mach Band Effect***.

Brightness Adaptation of Human Eye : Simultaneous Contrast



Simultaneous contrast. All small squares have exactly the same intensity but they appear progressively darker as background becomes lighter.

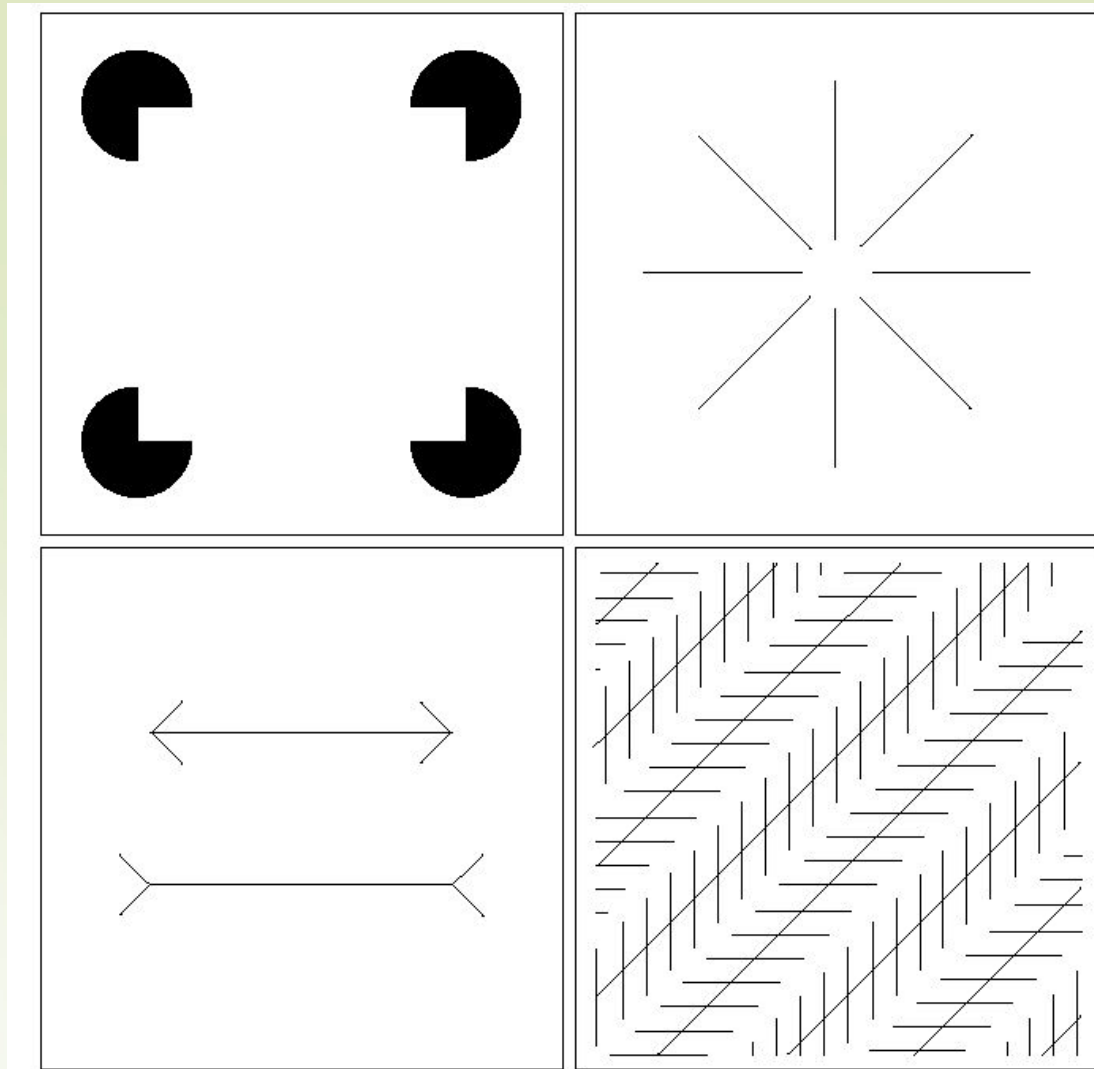
Simultaneous Contrast



a b c

FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

Optical illusion



Visible Spectrum

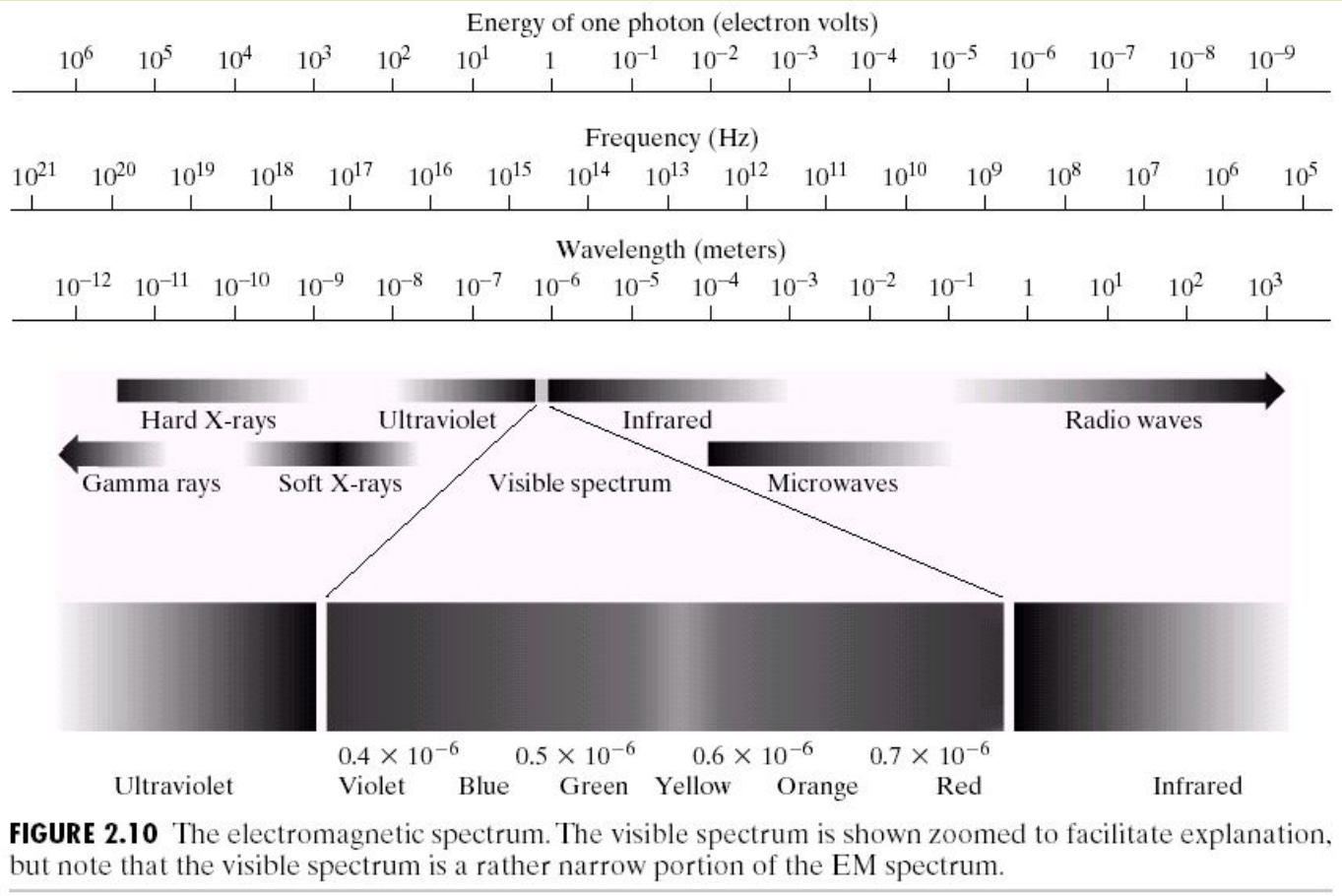
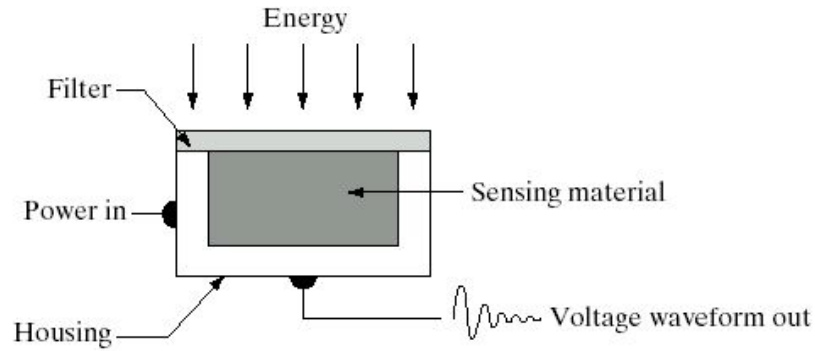


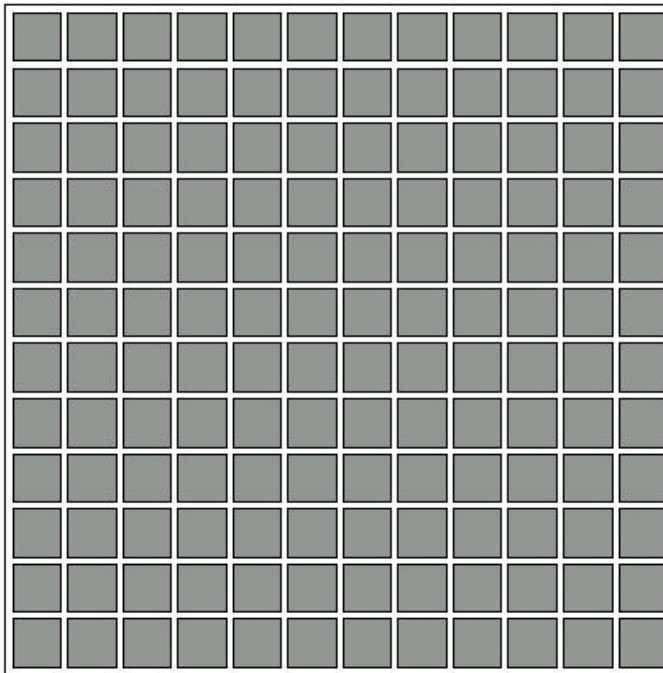
Image Sensors



Single sensor



Line sensor



Array sensor

Image Sensors : Single Sensor

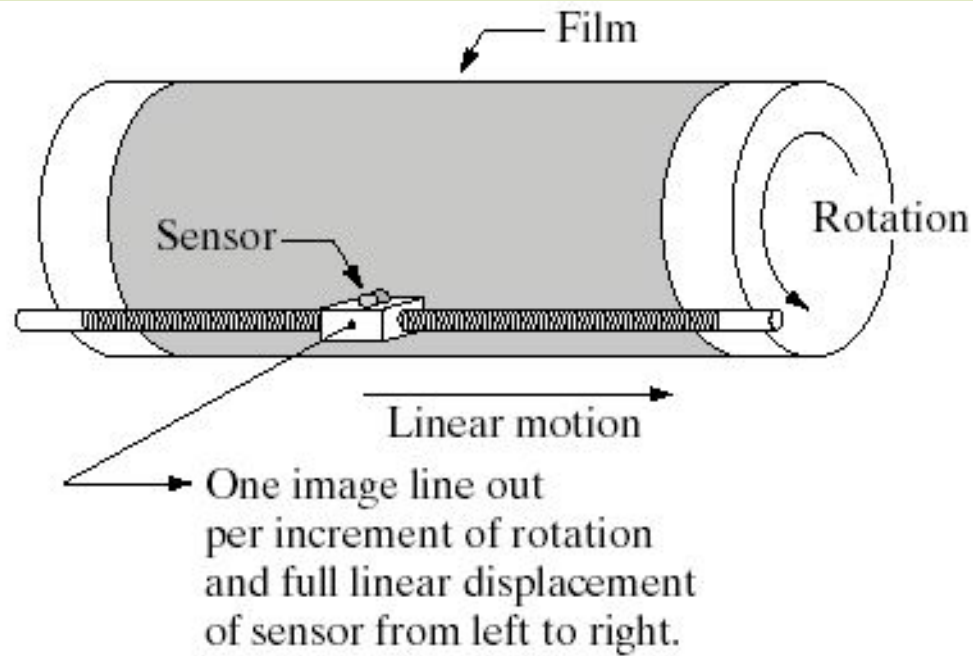
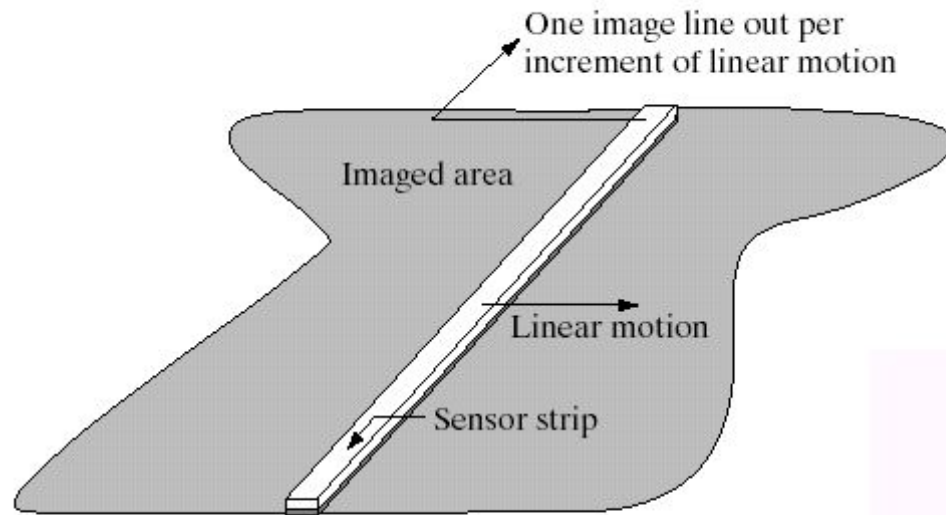
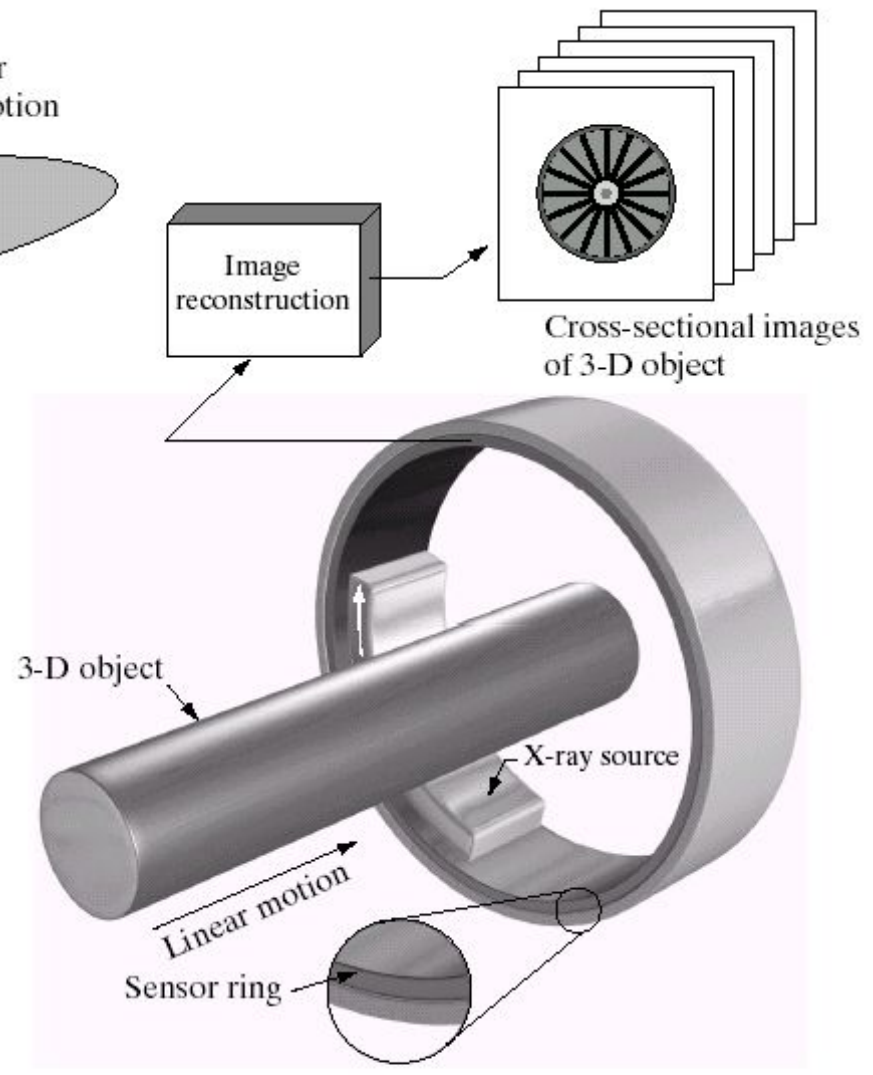


FIGURE 2.13 Combining a single sensor with motion to generate a 2-D image.

Image Sensors : Line Sensor



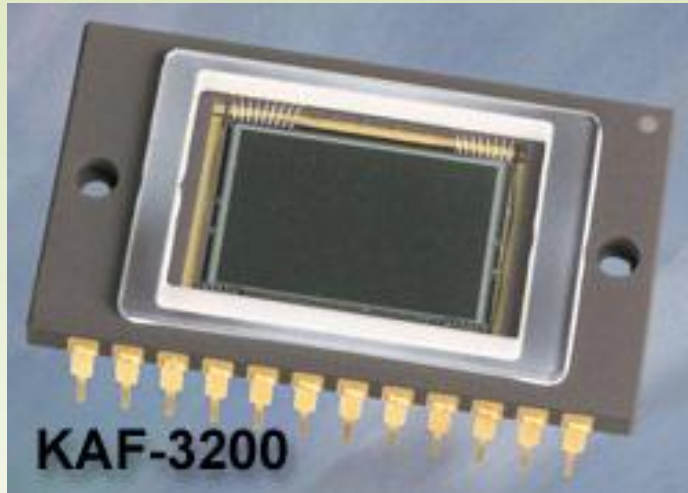
Fingerprint sweep sensor



Computerized Axial Tomography

Image Sensors : Array Sensor

Charge-Coupled Device (CCD)



CCD KAF-3200E from Kodak.
(2184 x 1472 pixels,
Pixel size 6.8 microns²)

- ♦ Used for convert a continuous image into a digital image
- ♦ Contains an array of light sensors
- ♦ Converts photon into electric charges accumulated in each sensor unit

Image Sensor: Inside Charge-Coupled Device

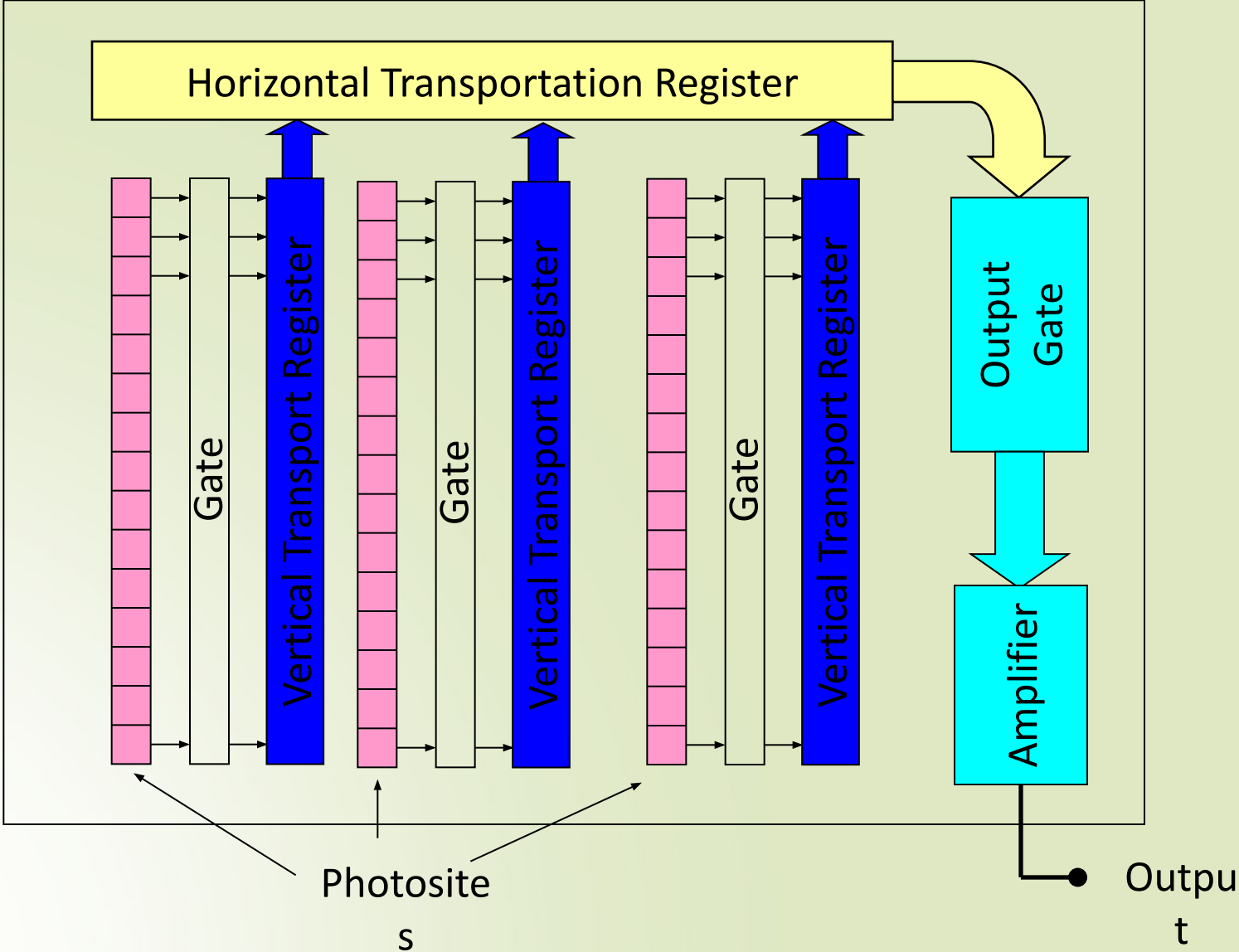
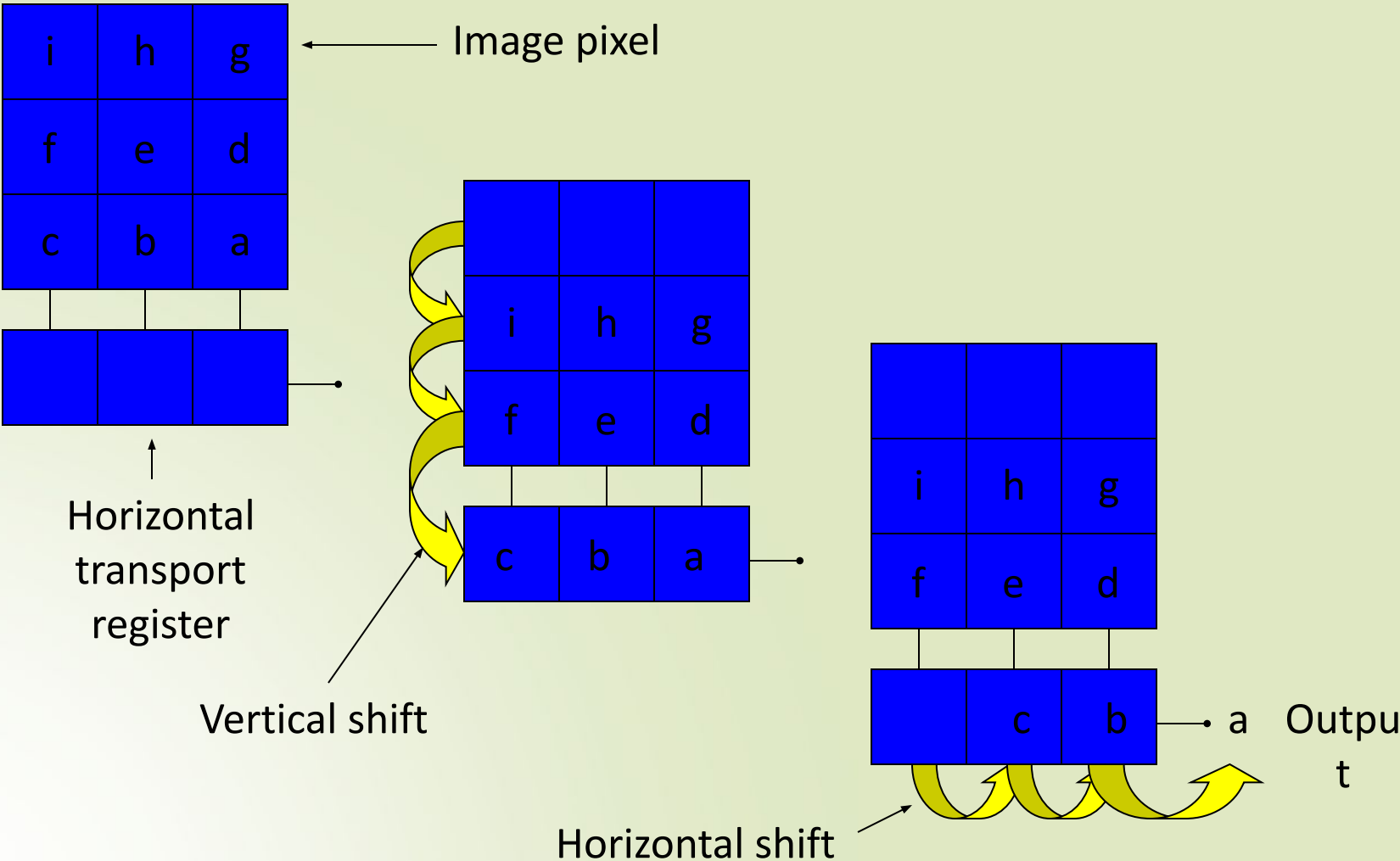


Image Sensor: How CCD works



Fundamentals of Digital Images



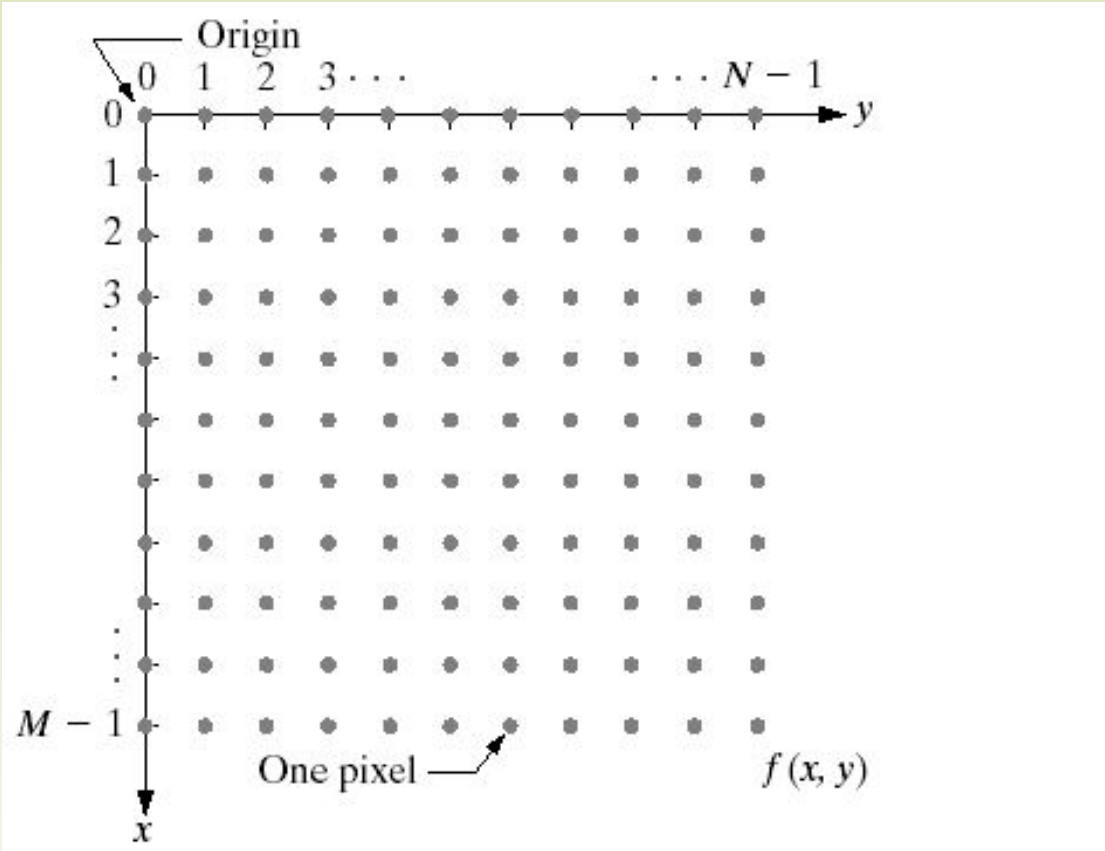
- ♦ An image: **a multidimensional function of spatial coordinates.**
- ♦ Spatial coordinate: (x,y) for 2D case such as photograph,
 (x,y,z) for 3D case such as CT scan images
 (x,y,t) for movies
- ♦ The function f may represent intensity (for monochrome images) or color (for color images) or other associated values.

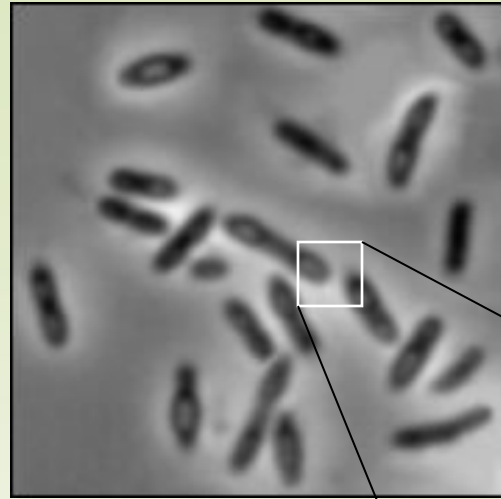
Digital Images

Digital image: an image that has been discretized both in spatial coordinates and associated value.

- ♦ Consist of 2 sets:(1) a point set and (2) a value set
- ♦ Can be represented in the form
$$I = \{(x, a(x)): x \in X, a(x) \in F\}$$
where X and F are a point set and value set, respectively.
- ♦ An element of the image, $(x, a(x))$ is called a **pixel** where
 - x is called the pixel location and
 - $a(x)$ is the pixel value at the location x

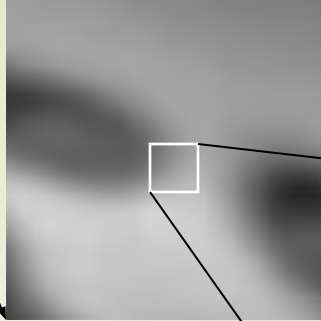
Conventional Coordinate for Image Representation



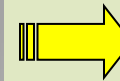


Intensity image or monochrome image

each pixel corresponds to light intensity normally represented in gray scale (gray level).

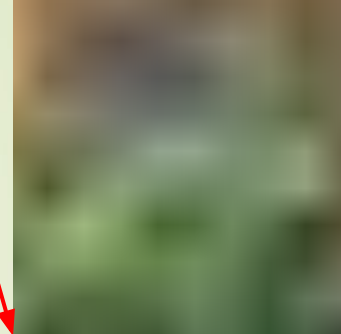


Gray scale values



10	10	16	28
9	6	26	37
15	25	13	22
32	15	87	39

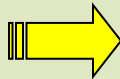
Digital Image Types : RGB Image



Color image or RGB image:

each pixel contains a vector representing red, green and blue components.

RGB components



10	10	16	28		
9	65	70	56	43	
15	32	99	70	56	78
32	21	60	90	96	67
	54	85	85	43	92
		32	65	87	99

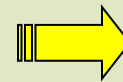
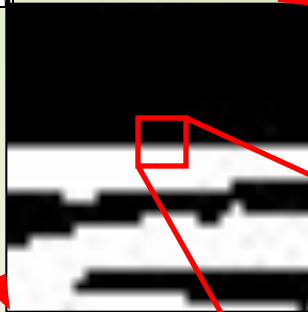
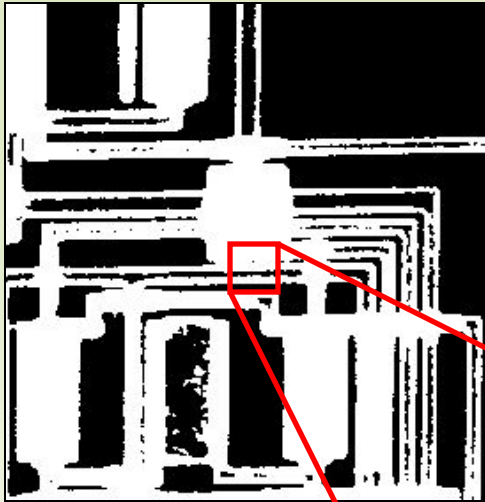
Image Types : Binary Image

Binary image or black and white image

Each pixel contains one bit :

1 represent white

0 represents black

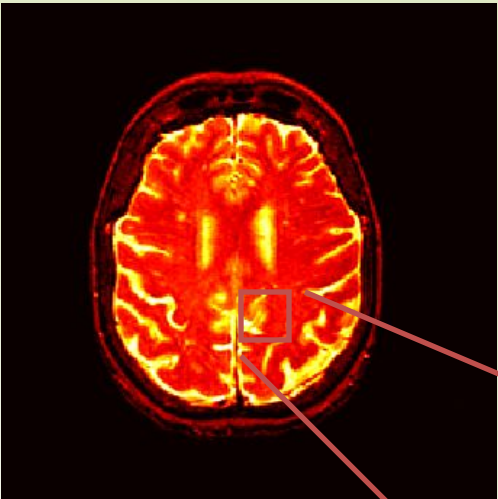


Binary data

0	0	0	0
0	0	0	0
1	1	1	1
1	1	1	1

Index image

Each pixel contains index number pointing to a color in a color table



1	4	9
6	4	7
6	5	2

Index value

Color Table

Index No.	Red component	Green component	Blue component
1	0.1	0.5	0.3
2	1.0	0.0	0.0
3	0.0	1.0	0.0
4	0.5	0.5	0.5
5	0.2	0.8	0.9
...	...	...	...

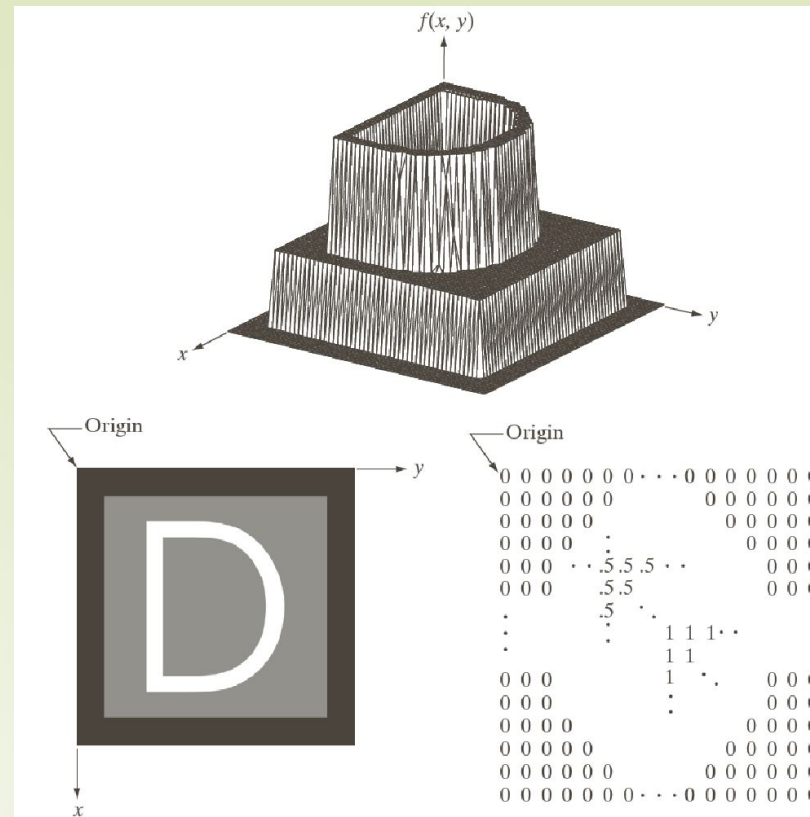
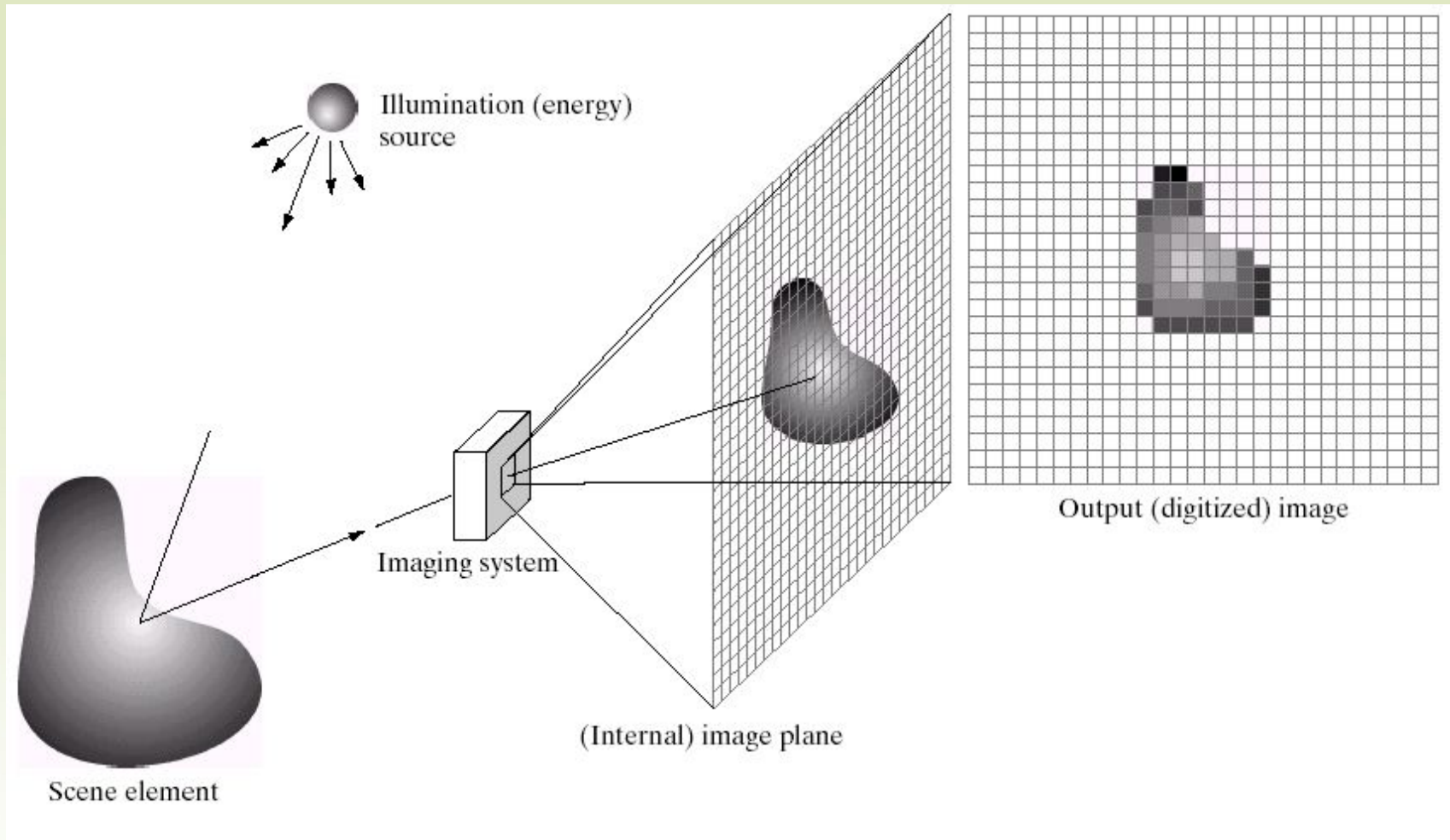
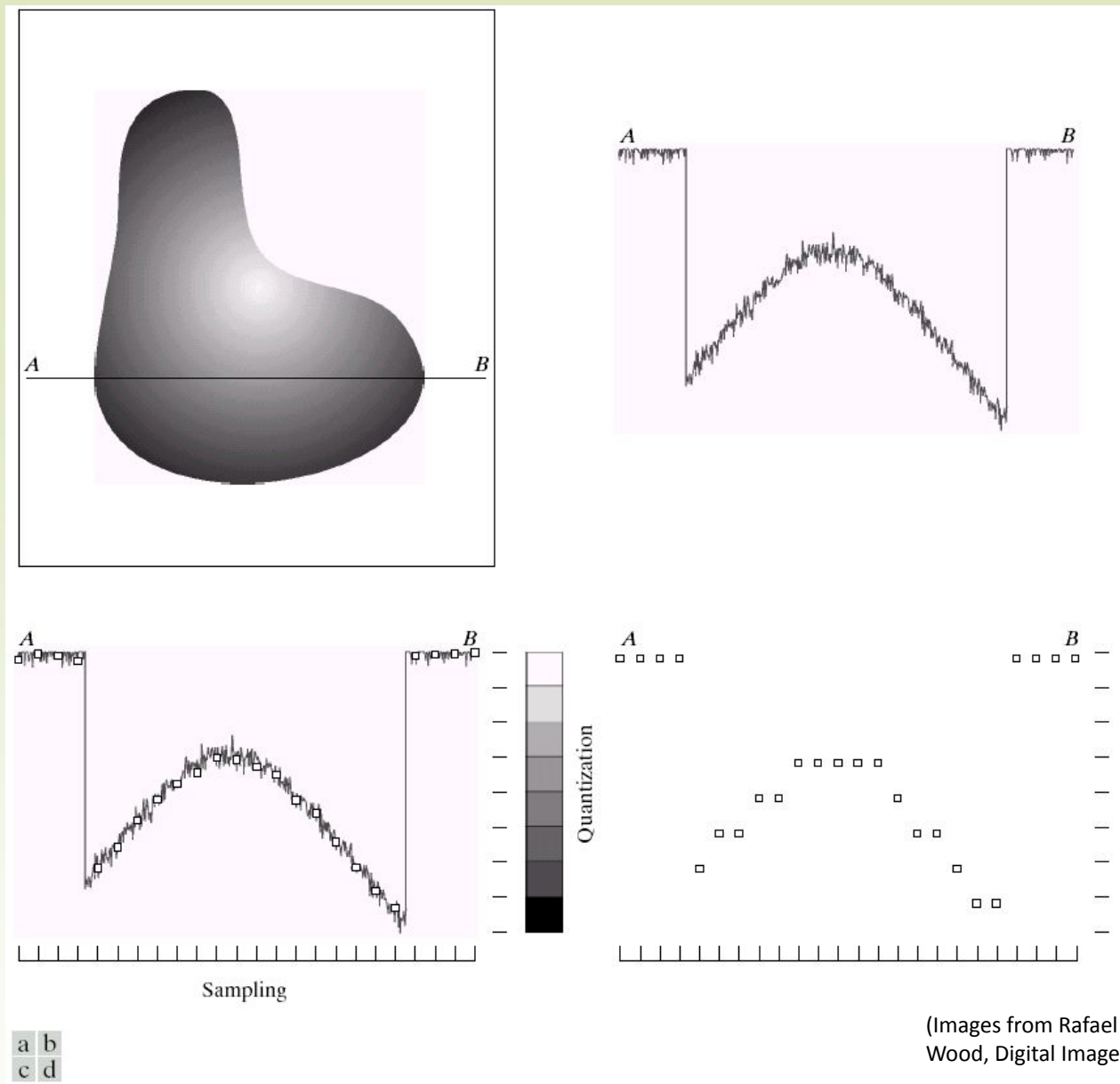


FIGURE 2.18
 (a) Image plotted as a surface.
 (b) Image displayed as a visual intensity array.
 (c) Image shown as a 2-D numerical array (0, .5, and 1 represent black, gray, and white, respectively).

Digital Image Acquisition Process



Generating a Digital Image



(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from A to B in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

Image Sampling and Quantization

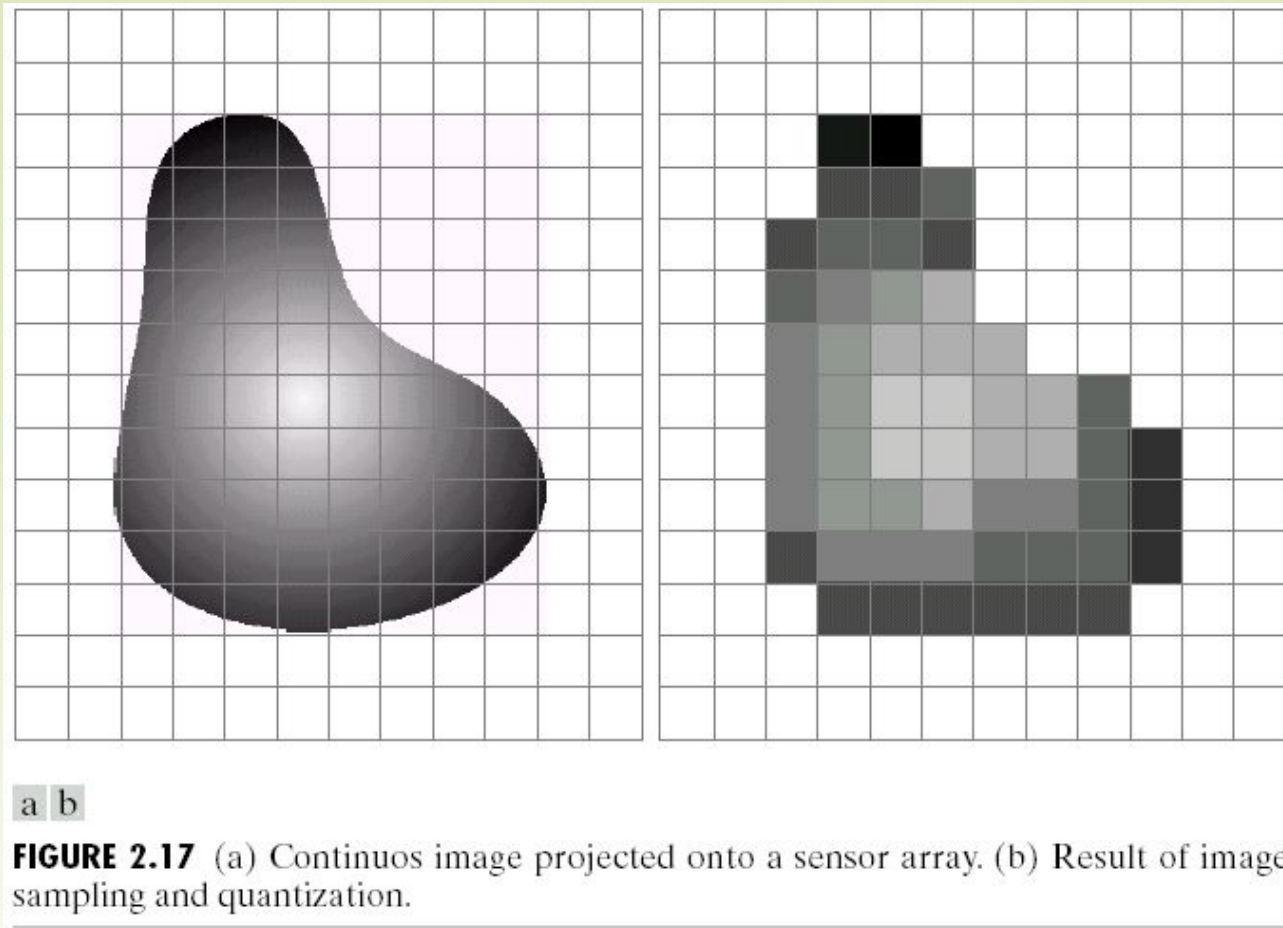
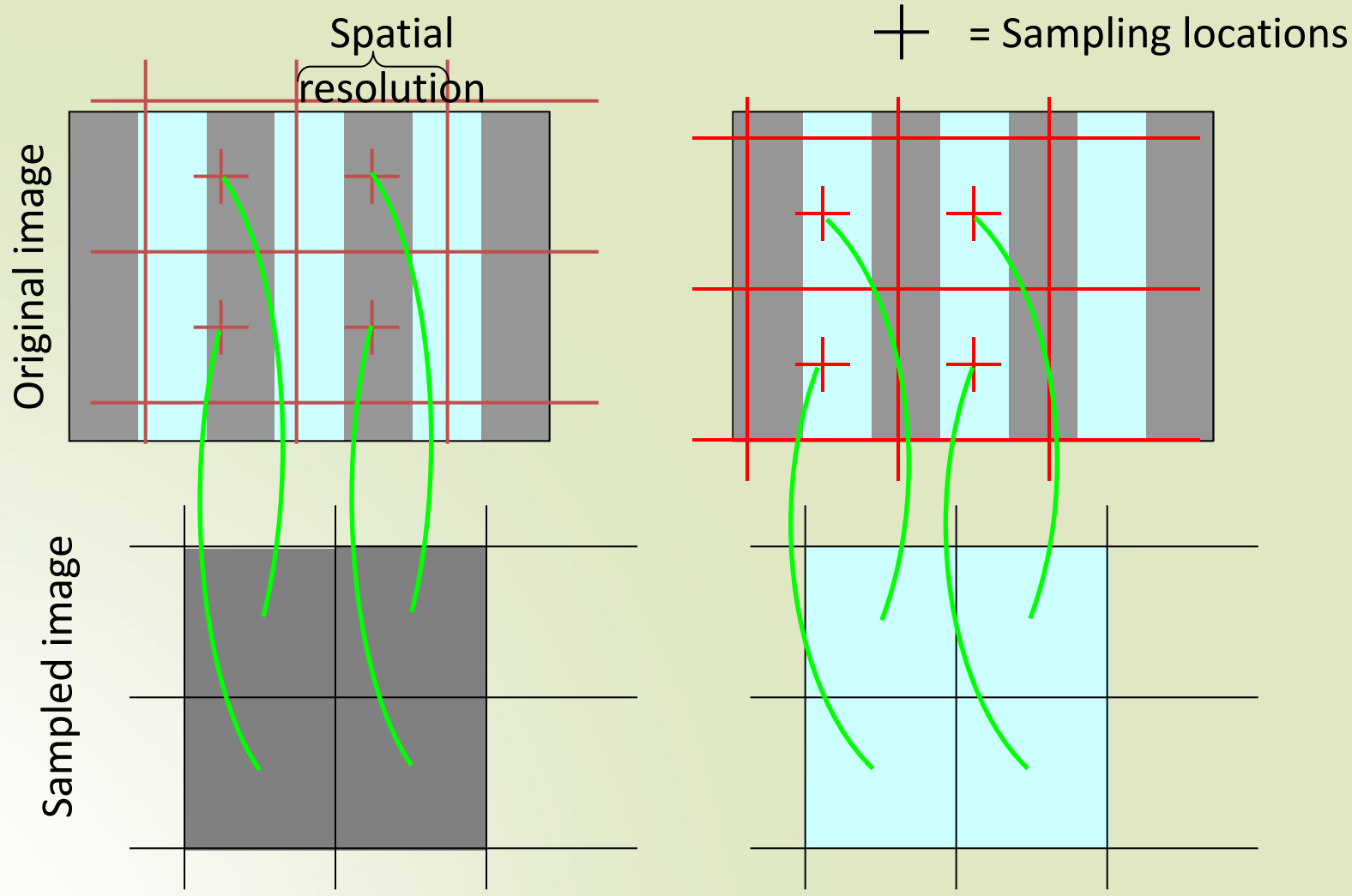


Image sampling: discretize an image in the spatial domain

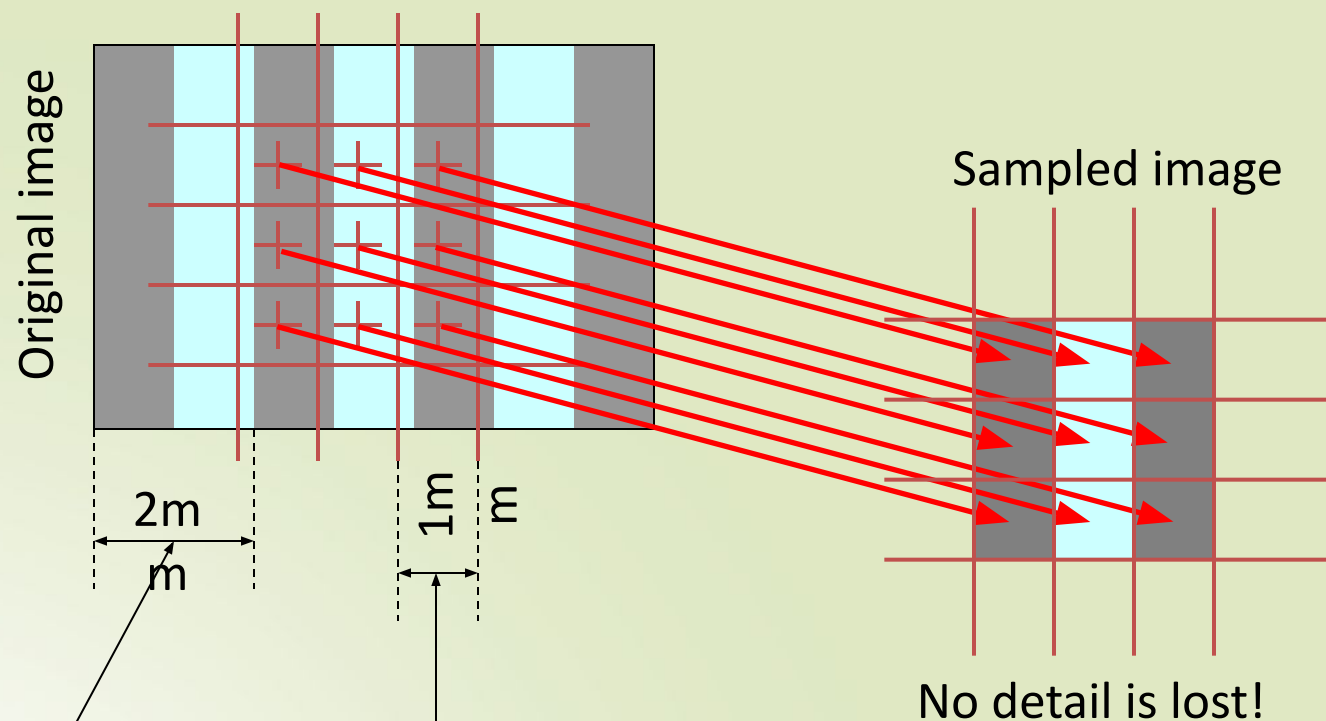
Spatial resolution / image resolution: pixel size or number of pixels

How to choose the spatial resolution



Under sampling, we lost some image details!

How to choose the spatial resolution : Nyquist Rate



Minimum
Period

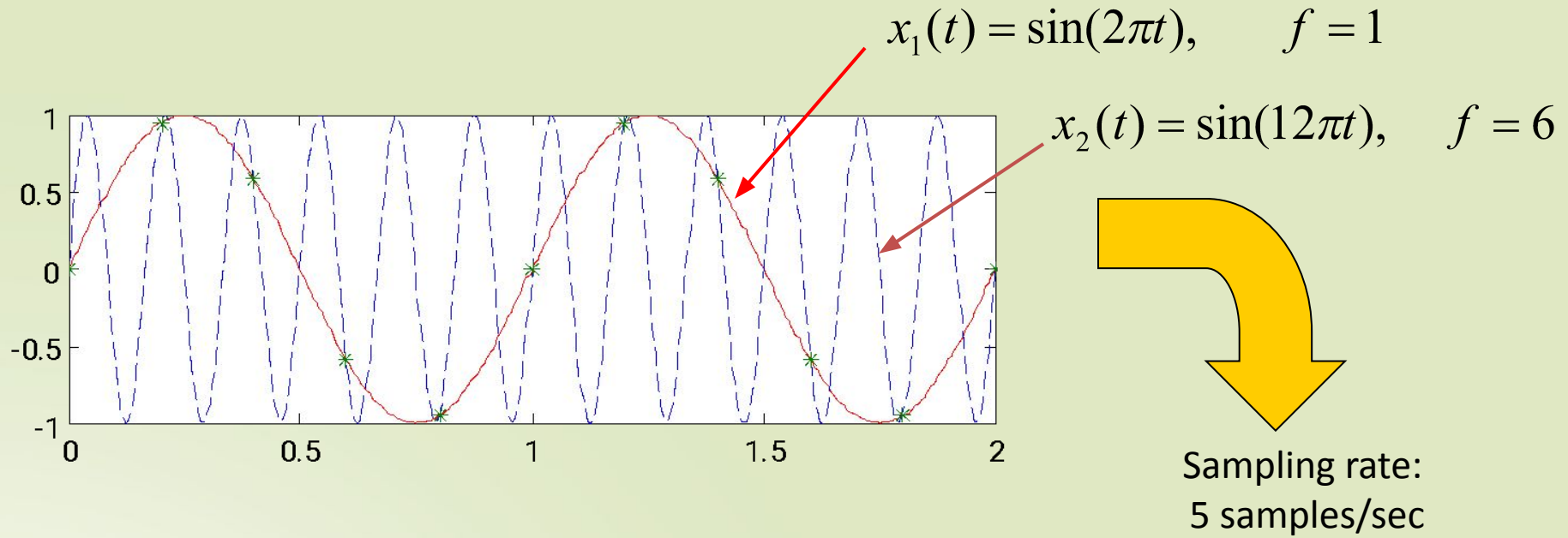
Spatial
resolution
(sampling rate)

⊕ = Sampling locations

Nyquist Rate:

Spatial resolution must be less or
equal
half of the minimum period of the
image
or sampling frequency must be greater
or

Aliased Frequency

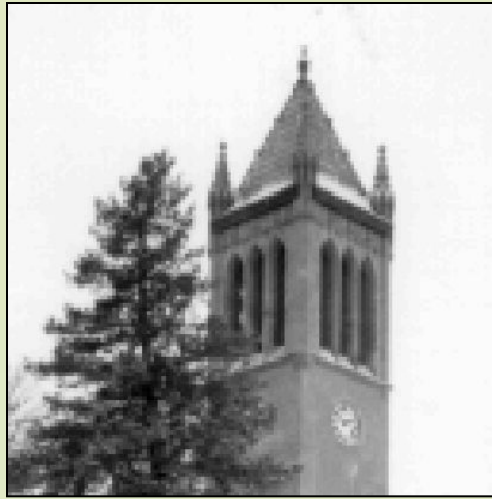


Two different frequencies but the same results !

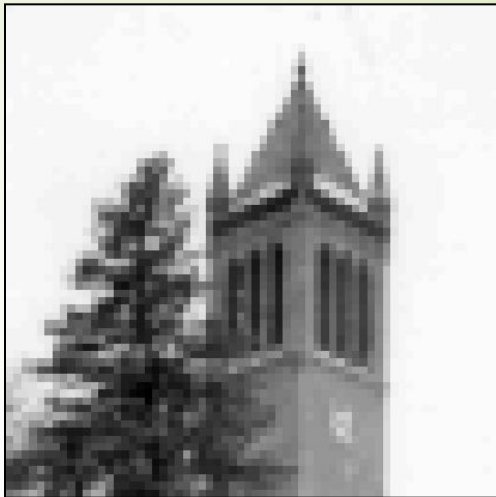
Effect of Spatial Resolution



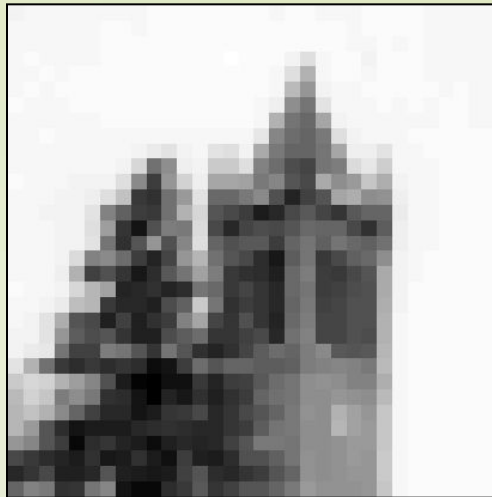
256x256 pixels



128x128 pixels



64x64 pixels



32x32 pixels

Effect of Spatial Resolution

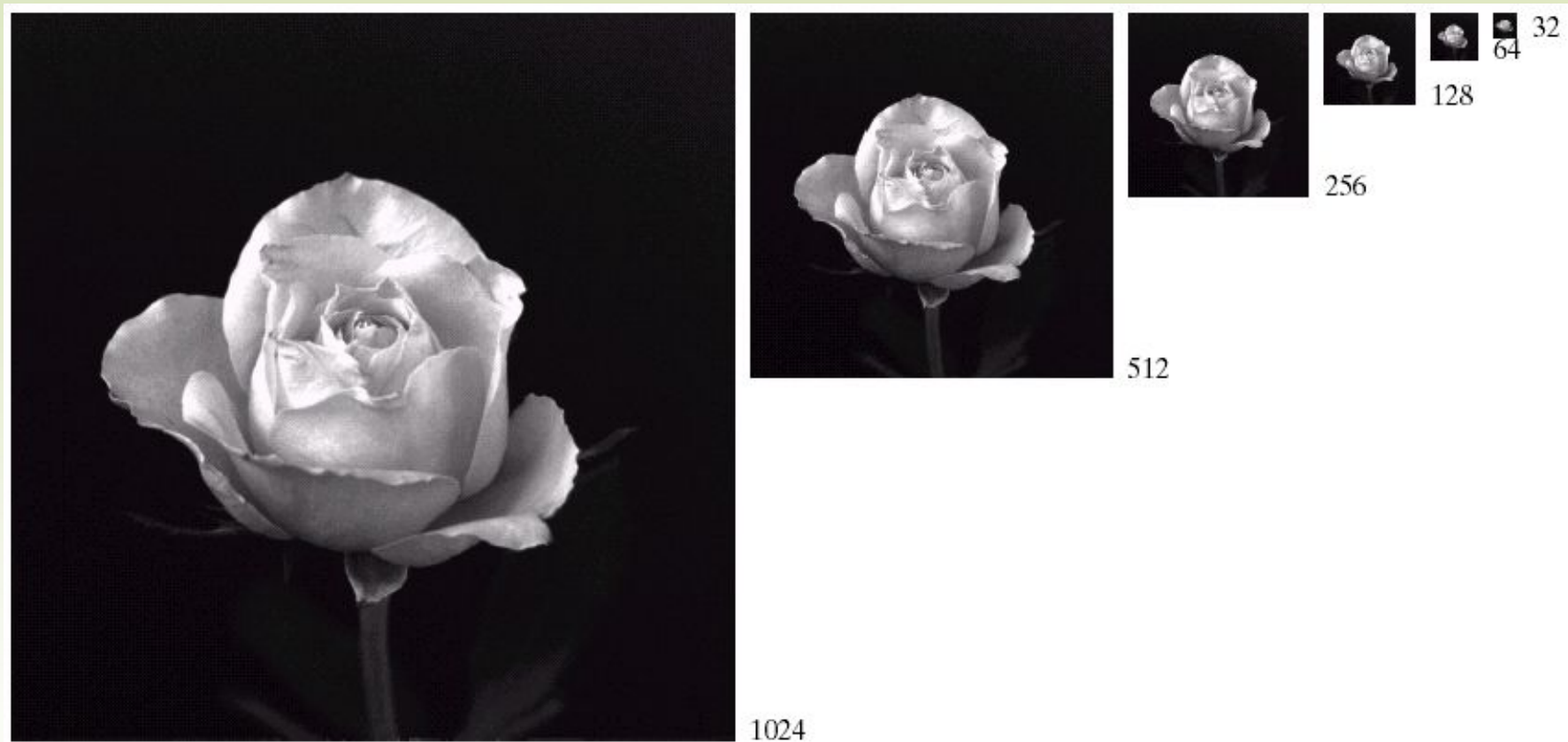
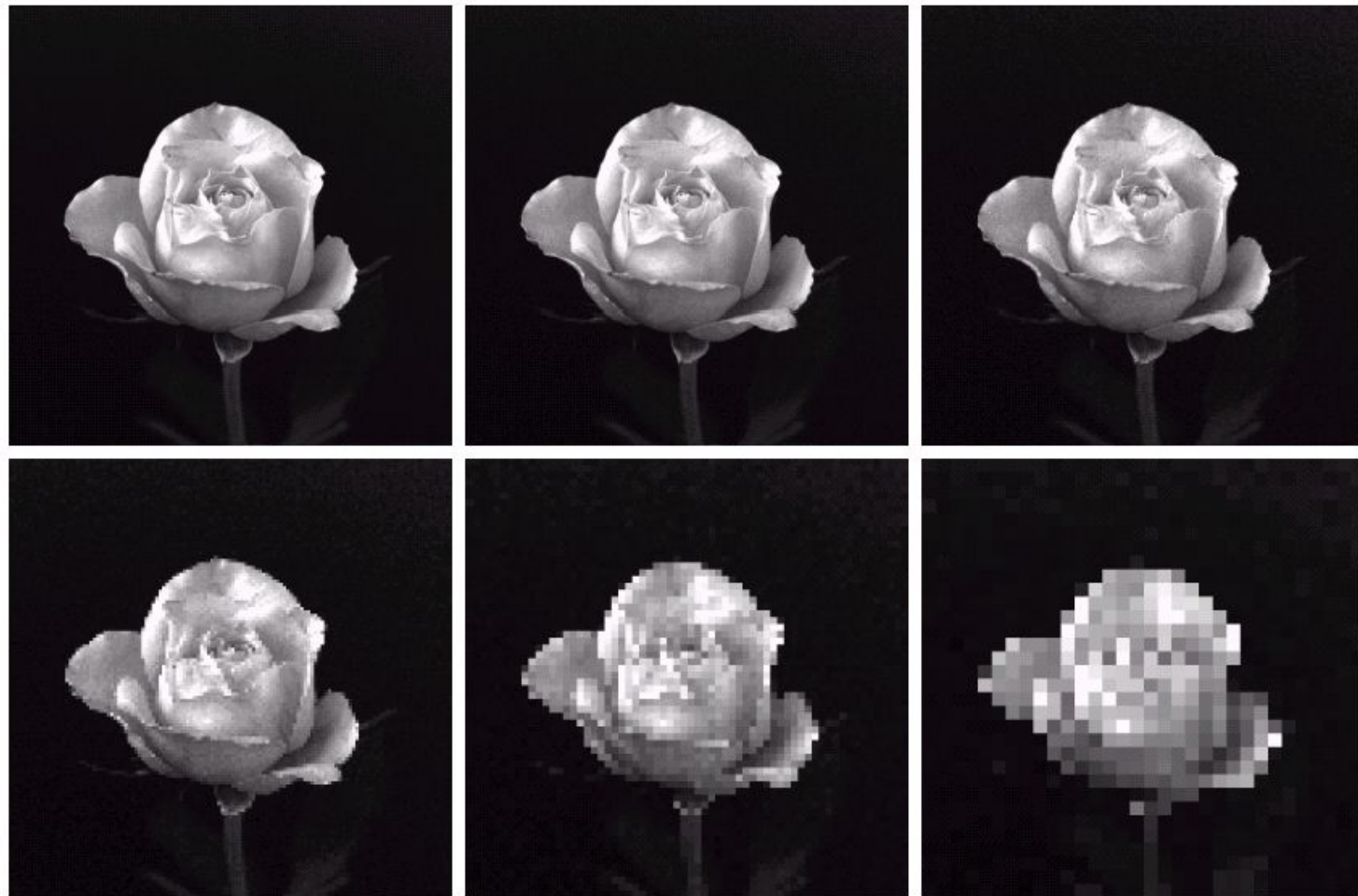


FIGURE 2.19 A 1024×1024 , 8-bit image subsampled down to size 32×32 pixels. The number of allowable gray levels was kept at 256.

Effect of Spatial Resolution



a	b	c
d	e	f

FIGURE 2.20 (a) 1024×1024 , 8-bit image. (b) 512×512 image resampled into 1024×1024 pixels by row and column duplication. (c) through (f) 256×256 , 128×128 , 64×64 , and 32×32 images resampled into 1024×1024 pixels.

Can we increase spatial resolution by interpolation ?

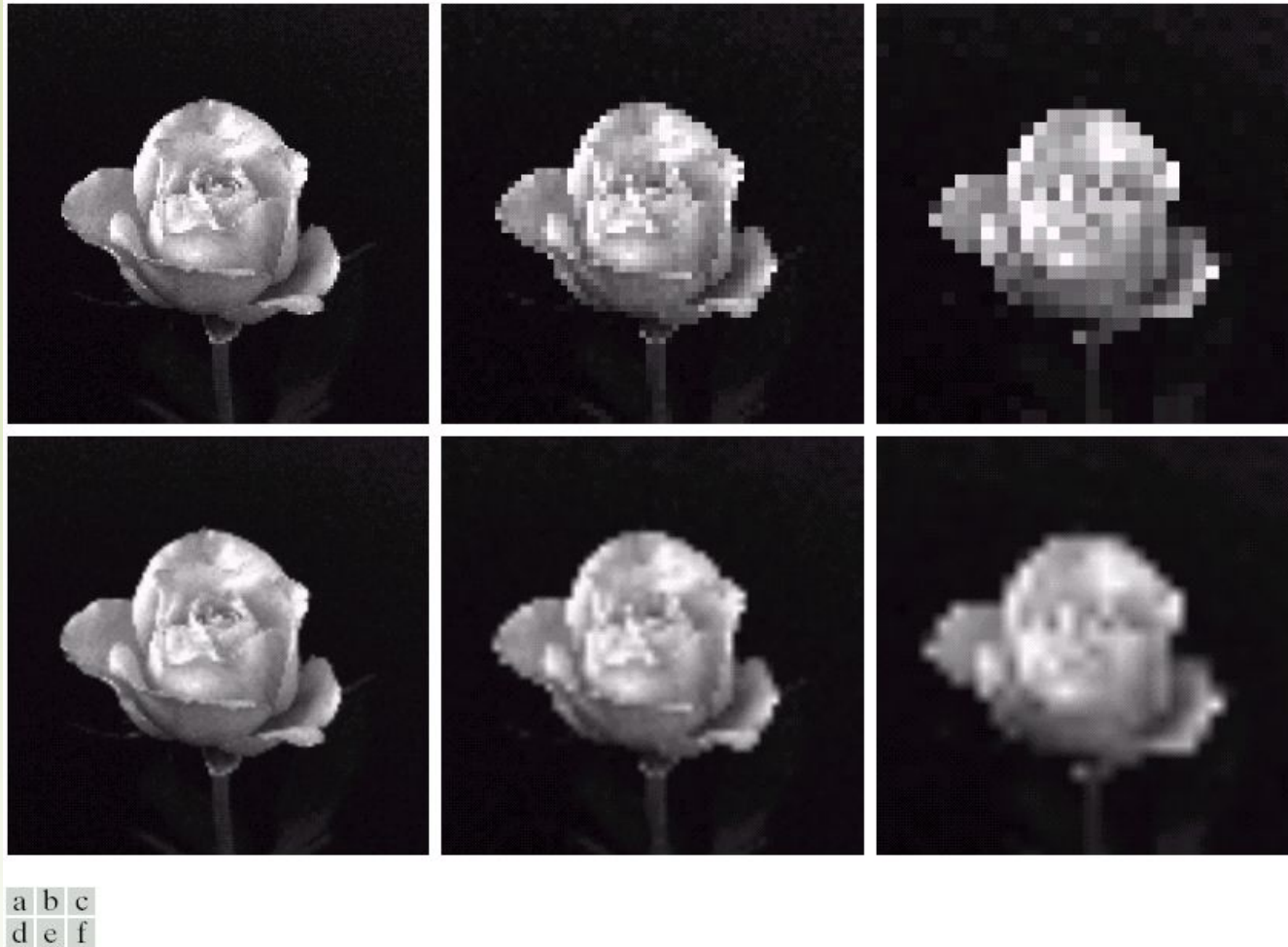


FIGURE 2.25 Top row: images zoomed from 128×128 , 64×64 , and 32×32 pixels to 1024×1024 pixels, using nearest neighbor gray-level interpolation. Bottom row: same sequence, but using bilinear interpolation.

Down sampling is an irreversible process.

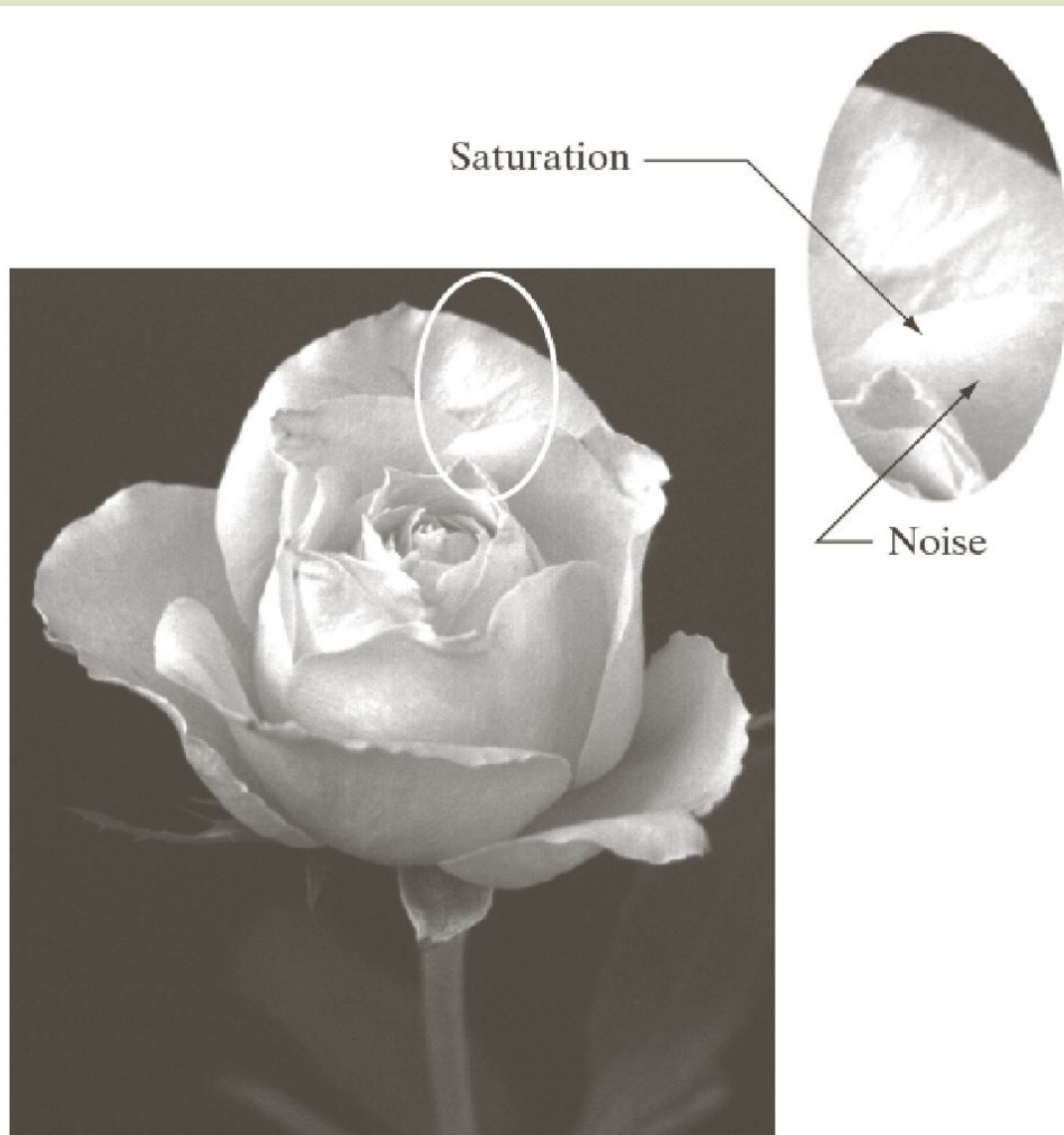


FIGURE 2.19 An image exhibiting saturation and noise. Saturation is the highest value beyond which all intensity levels are clipped (note how the entire saturated area has a high, *constant* intensity level). Noise in this case appears as a grainy texture pattern. Noise, especially in the darker regions of an image (e.g., the stem of the rose) masks the lowest detectable true intensity level.

TABLE 2.1Number of storage bits for various values of N and k .

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Image Quantization

Image quantization:

discretize continuous pixel values into discrete numbers

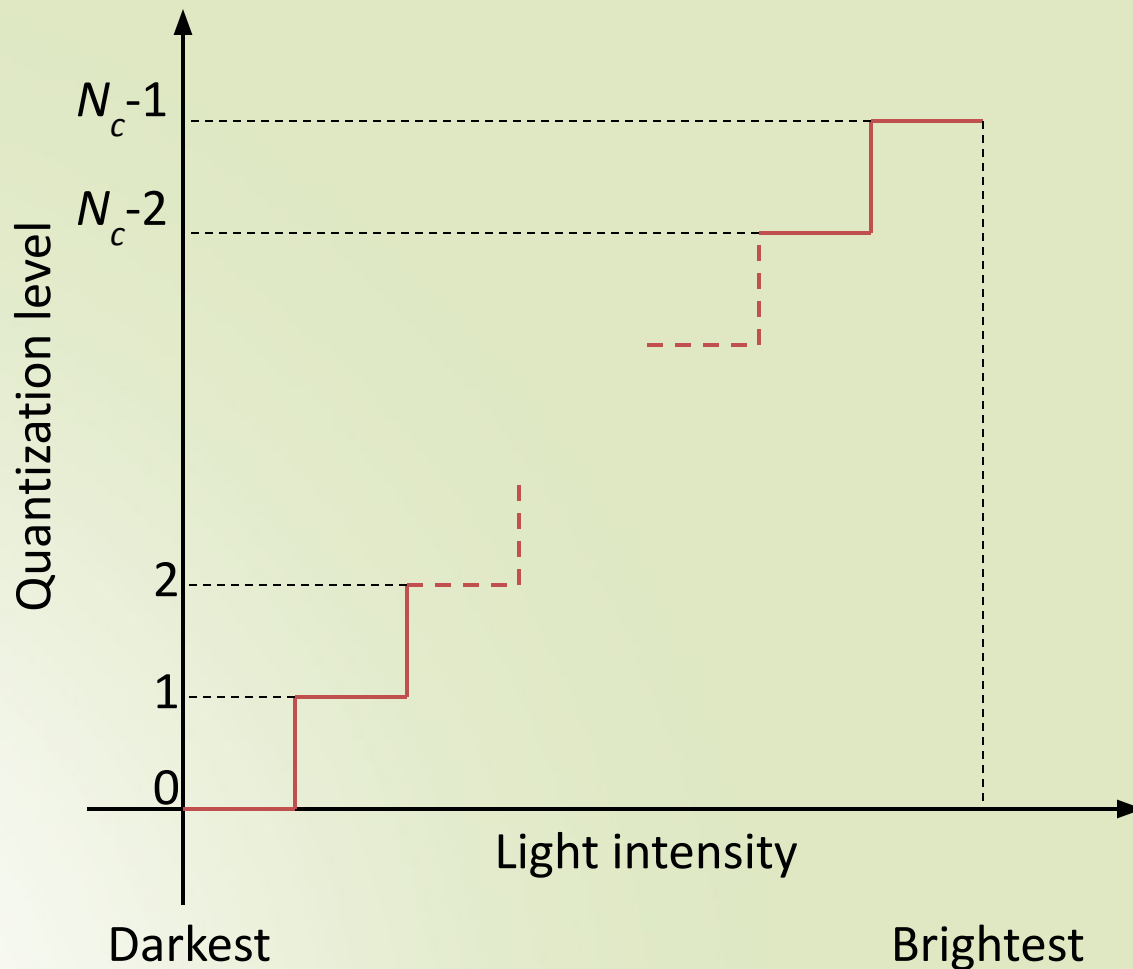
Color resolution/ color depth/ levels:

- No. of colors or gray levels or
- No. of bits representing each pixel value
- No. of colors or gray levels N_c is given by

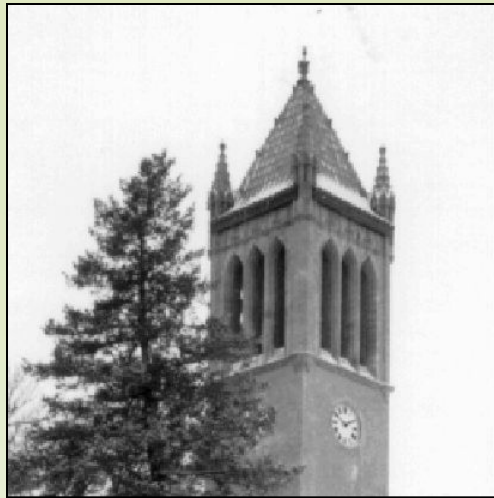
$$N_c = 2^b$$

where b = no. of bits

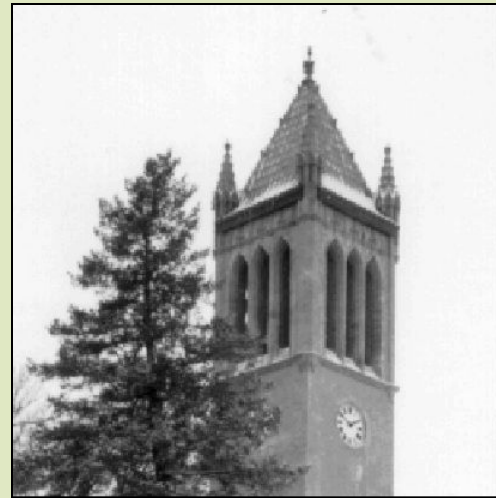
Quantization function



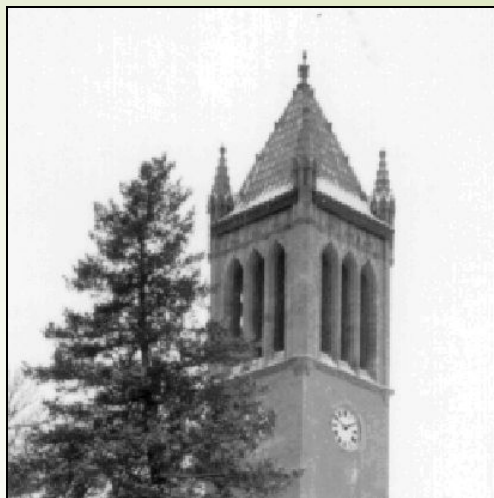
Effect of Quantization Levels



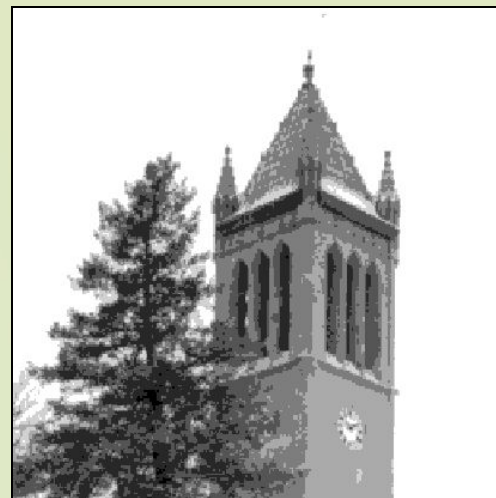
256 levels



128 levels

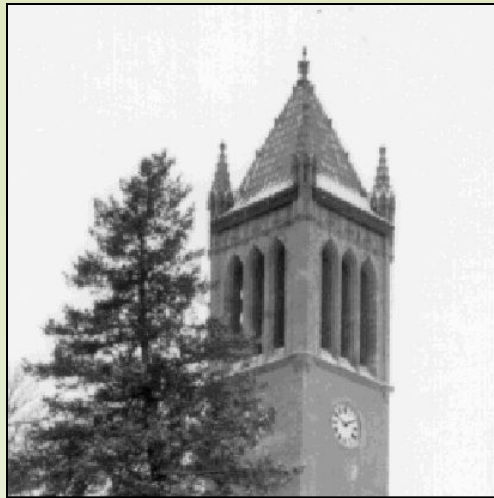


64 levels

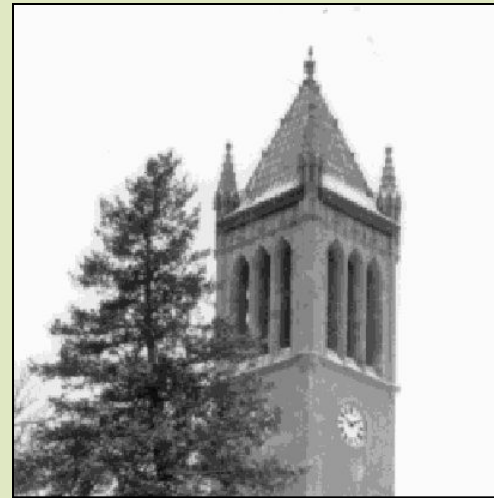


32 levels

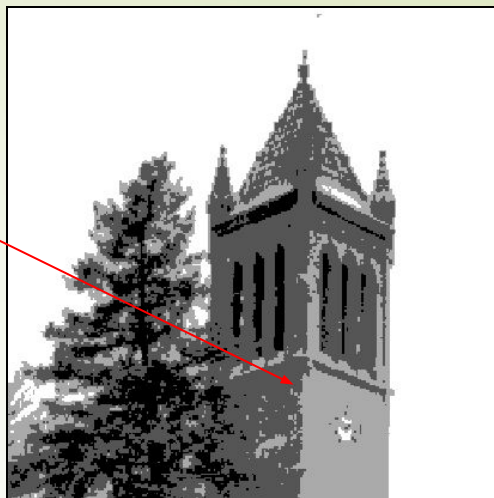
Effect of Quantization Levels (cont.)



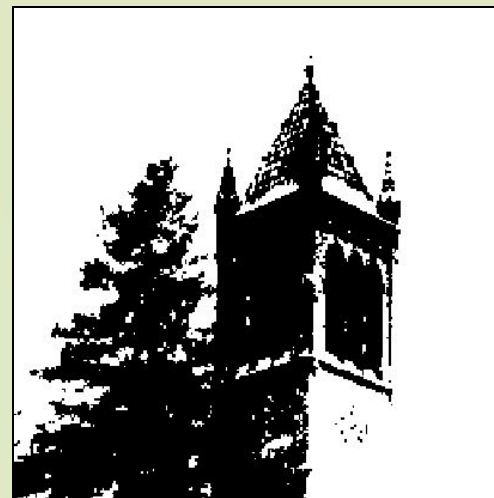
16 levels



8 levels



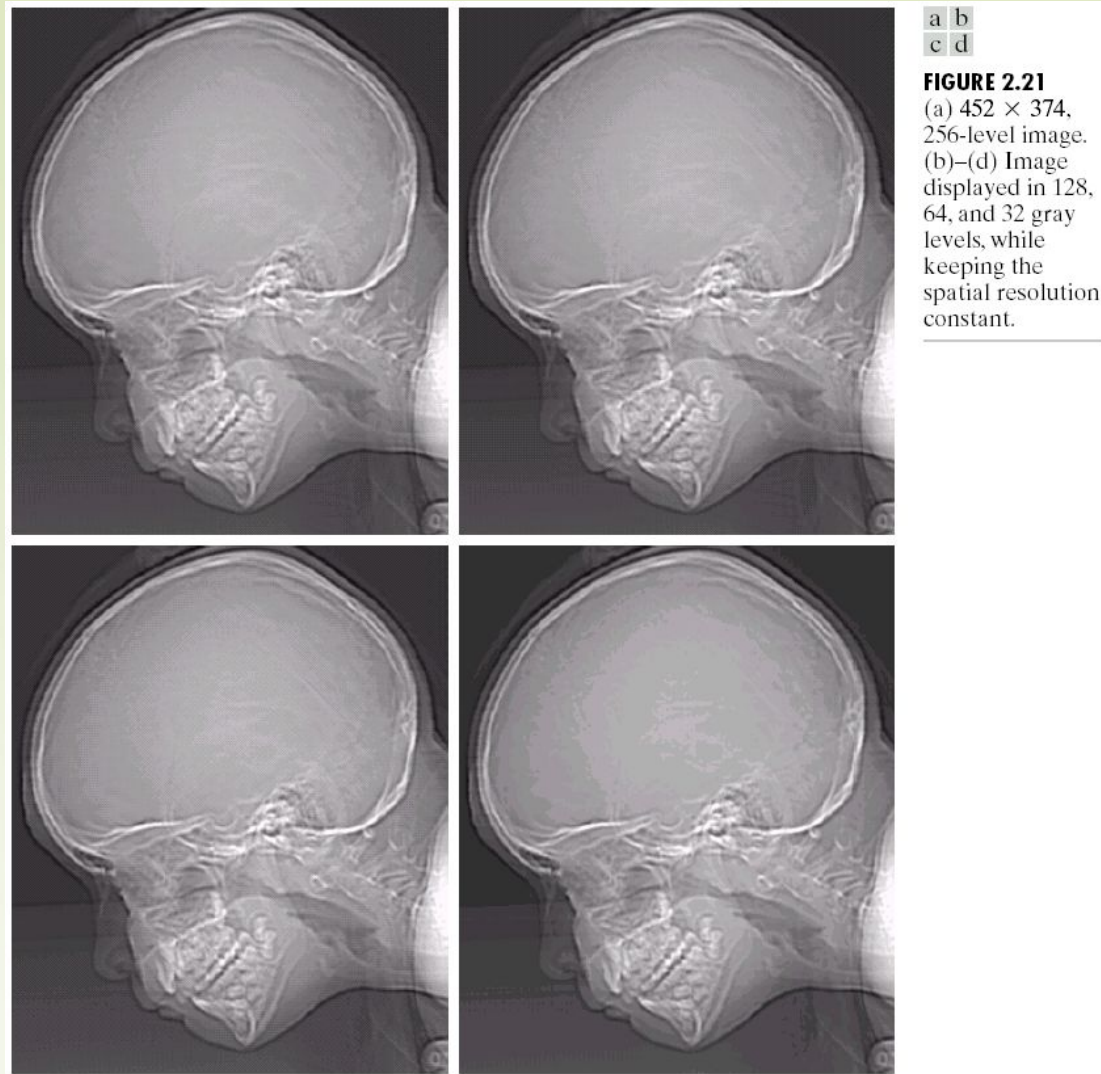
4 levels



2 levels

In this image,
it is easy to see
false contour.

Chapter 2: Digital Image Fundamentals



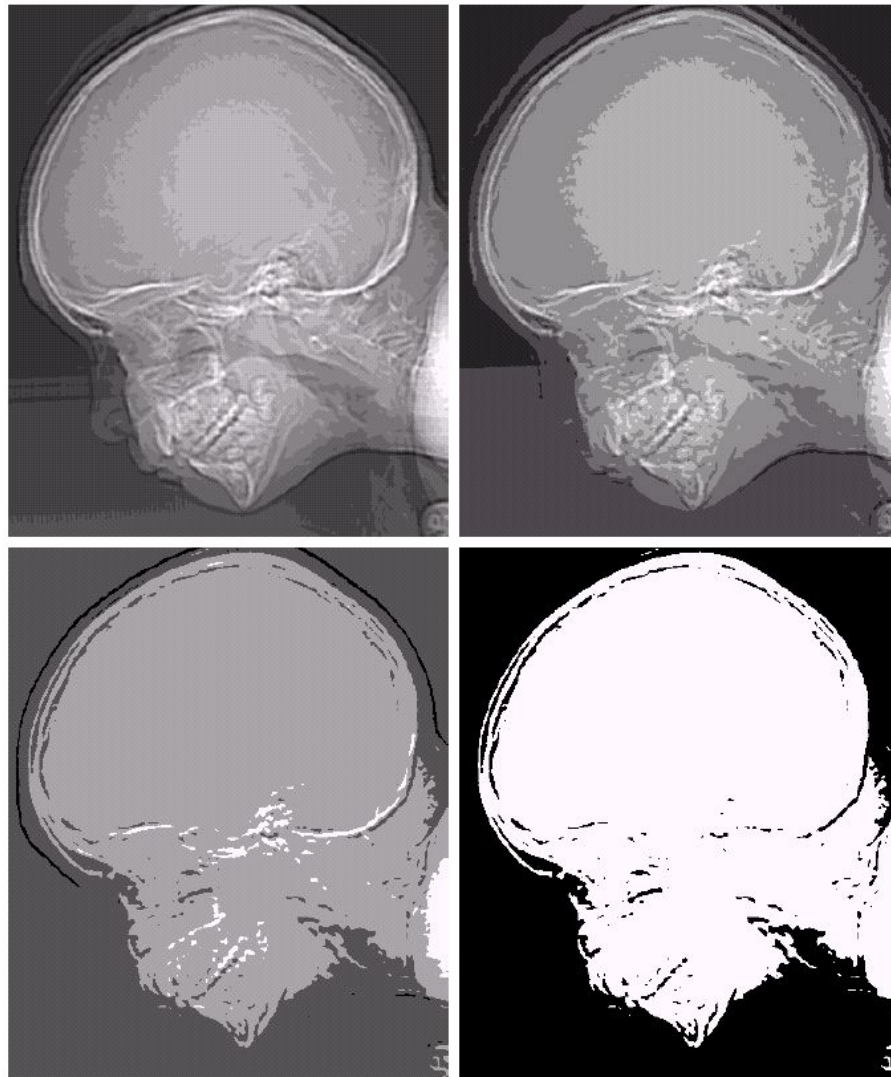
Chapter 2: Digital Image Fundamentals

e f
g h

FIGURE 2.21

(Continued)

(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)



How to select the suitable size and pixel depth of images

The word “suitable” is subjective: depending on “subject”.



Low detail image

Lena image



Medium detail image

Cameraman image



High detail image

To satisfy human mind

1. For images of the same size, the low detail image may need more pixel depth.
2. As an image size increase, fewer gray levels may be needed.

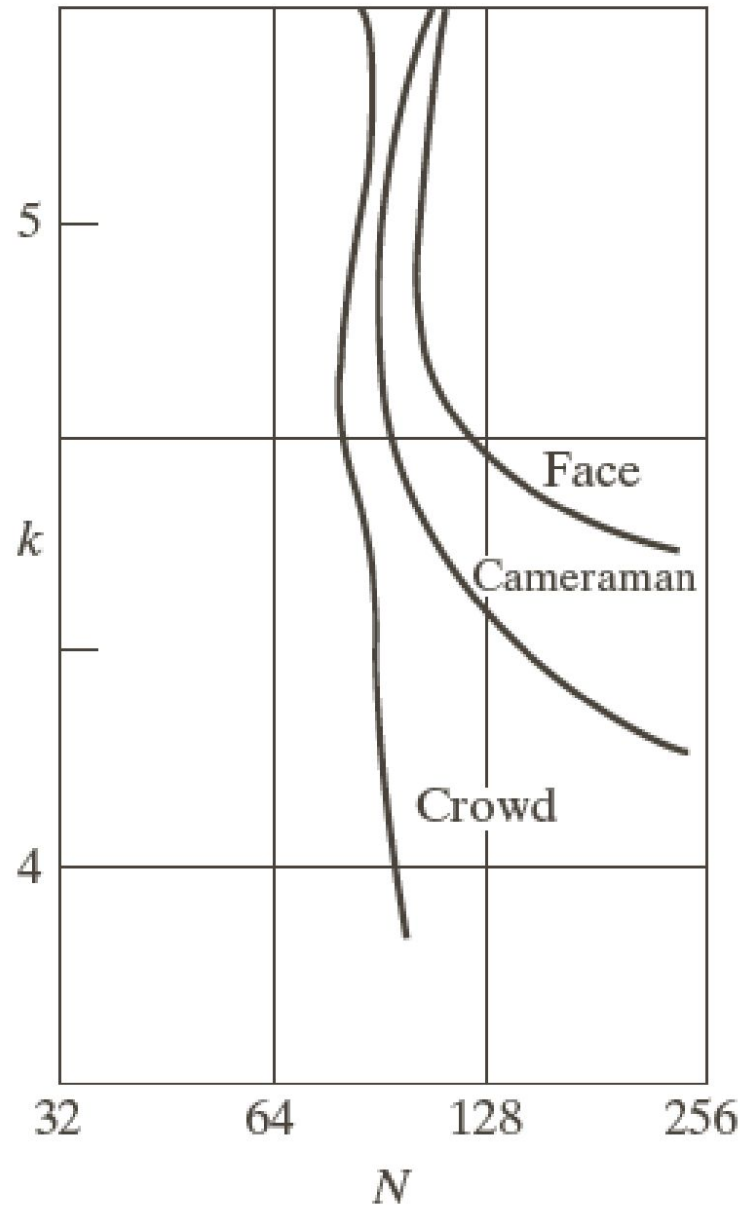


FIGURE 2.23

Typical isopreference curves for the three types of images in Fig. 2.22.

Resizing Images

Zooming : Creating new pixel locations
 Assigning gray-level values to these locations

Solution: Interpolation

**Nearest
Neighbor**



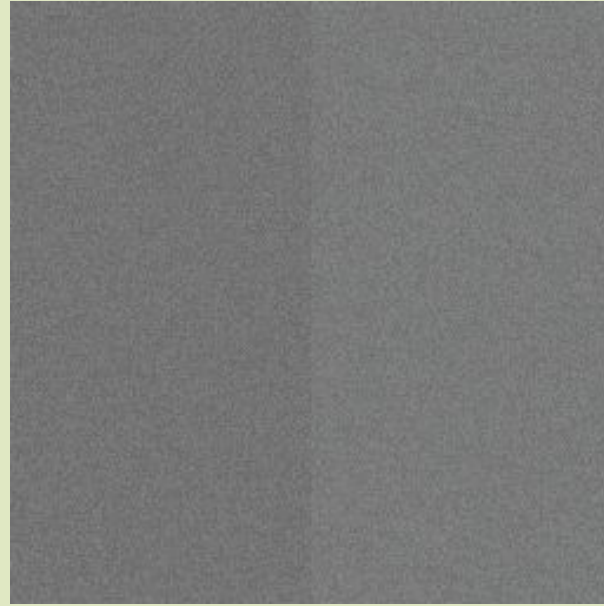
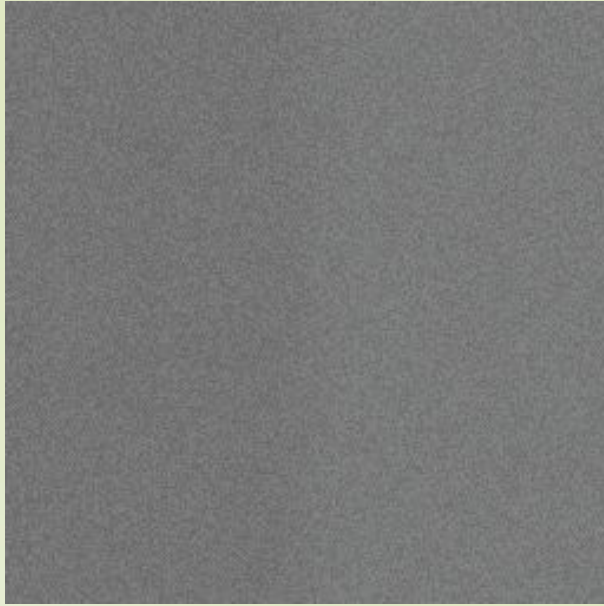
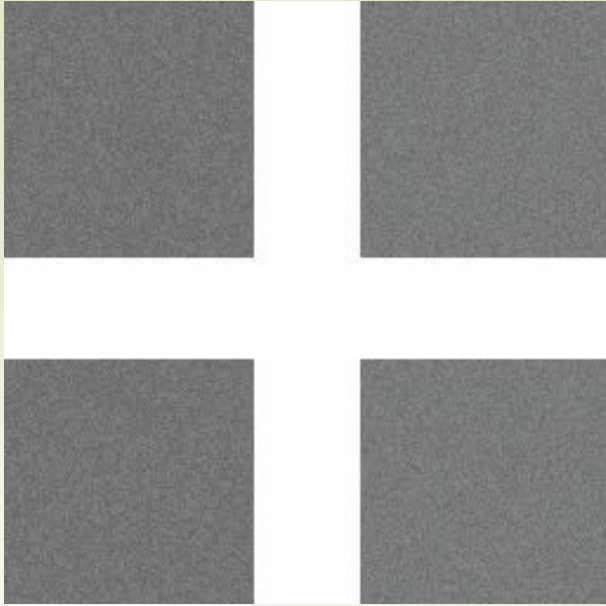
Bilinear



128 → 1024

64 → 1024

Human vision: Distinguish ability for Difference in brightness



Regions with 5% brightness difference

Acknowledgment

- “Digital Image Processing”, 2/ E, Rafael C. Gonzalez & Richard E. Woods,



Questions?